T2K 4.5×10^{20}-POT 3-Flavour Electron-Antineutrino Appearance Analysis

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Abstract

We report the results of the 3-flavour $\bar{\nu}_e$ appearance analysis, performed by the *VALOR* group, on the combined Run 5+6 dataset, using only data taken in antineutrino beam mode. This analysis adapts the work presented in Refs. [1], [2], [3], [4], [5], [6], [7], [8] and [9] to this dataset. The Run 5+6 dataset corresponds to an integrated J-PARC neutrino beam exposure of 4.5×10^{20} POT.

This analysis predicts 4.183 single e-like ring events in Super K in the absence of any oscillation and XXX were observed.

A $\bar{\nu}_e$-appearance analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter.

The observed reconstructed energy spectrum of single e-like ring events was compared, to the expected energy spectrum, for different signal hypotheses: absence and presence of $\bar{\nu}_e$ appearance consistent with the current best knowledge of the PMNS matrix, i.e. T2K+reactor joint fit with the Run1-4 data set.

A new not physical parameter, β, was introduced to define the absence or the presence of $\bar{\nu}_e$ appearance. All the 56 systematic parameters plus the oscillation parameters were marginalized, fixing β either to 0 or 1. The p-value given the null $\bar{\nu}_e$ appearance hypothesis was quoted performing many toy experiments and is XXX.

1. Introduction

In this technical note, we report the results of a 3-flavour $\bar{\nu}_e$-appearance analysis, performed by the *VALOR* group, on the combined Run 5+6 dataset, using only data taken in antineutrino beam mode (4.5×10^{20} POT). This dataset was collected in two run periods: May - June 2014 (Run 5, 0.506×10^{20} POT in antineutrino beam mode) and November 2014 - June 2015 (Run 6, XXX$\times10^{20}$ POT in antineutrino beam mode). This analysis builds on our other analyses presented in Refs. [1], [1], [2], [3], [4], [5], [6], [7], [8] and [9] , and adapts these analyses to antineutrino beam mode data. The main difference is that in the past a fully frequentist analysis was performed. Now a hybrid frequentist-bayesian method [46], which consists of marginalizing instead of fitting the systematic parameters as well as the mass hierarchy, is used. Prior distributions obtained from the T2K+reactor joint fit [36] were used in the analysis.

The goal of this analysis is to search for $\bar{\nu}_e$ appearance in the PMNS matrix contest and not to search for any physics beyond the Standard Model. For this reason, differently from the $\bar{\nu}_\mu$-disappearance analysis [9] common oscillation parameters are used for ν and $\bar{\nu}$. The consistency of the data with the expectation is given by comparing the predicted reconstructed energy spectrum of 1 e-like ring events at SuperK to the observed spectrum. The method of calculation of this predicted spectrum is presented in Sec. 2. This analysis uses flux tuning based on recent NA61 data [15][38]. It also uses an indirect extrapolation from ND280 to SuperK, which is implemented via a fit of flux and cross-section parameters to ND280 measurements of the muon momentum-angle distributions of $\bar{\nu}_\mu$ and ν_μ CC0π-like, CC1π-like and CC other events whose vertices are in FGD1[26]. Updated nominal predictions of the reconstructed energy spectrum of single e-like ring events in SuperK are presented for Run 5+6 in Sec. 2

The oscillation analysis is performed using neutrino oscillation probabilities computed in a 3-flavour framework including constant-density matter effects. Details of our oscillation probability calculations, estimates of the numerical accuracy of our calculations and comparisons with calculations used by the alternative T2K oscillation analyses are given in Appendices A and B of Ref. [4].

As in our previous analyses, the oscillation fit is made using the likelihood-ratio method [43]. The mathematical details of this method are given in Sec. 3.2 in Ref. [1]. We consider a total of 56 energy scale, neutrino flux, neutrino cross-section, efficiency, initial-state nuclear environment and final-state rescattering systematics plus the oscillation parameters (nuisance parameters). In this Run 5+6 analysis, all the nuisance parameters are marginalized [46] . The nuisance parameters and their effects on the extrapolated single e-like ring event reconstructed energy spectrum are described in Sec. 3. The analysis method is presented in detail in Sec. 4

The significance to $\bar{\nu}_e$ appearance was given as a p-value. Two different test statistics were used: the total number of observed events (N_{obs}), without using any information from the reconstructed energy shape, was used to quote the rate-only p-value, while $\Delta\chi^2 = \chi(\beta = 0) - \chi(\beta = 1)$ was used to quote the rate+shape p-value. More details are given in 4.3

The results of the measurement using the Run 5+6 dataset is given in Sec. 6.1 of this note.

Finally, a summary is given in Sec. 7.

2. Predictions of single *e*-like ring event reconstructed energy spectrum in SuperK

2.1. Methodology

In this analysis, the parameter of interest β which reweghts the $\bar{\nu}_e$ appearance signal component is kept fixed to either 0 (no appearance) or 1 (appearance consistent with the current best knowledge of the PMNS matrix given by the T2K+reactor Run1-4 joint fit result). The reconstructed energy (E_r) spectrum of single e-like ring events observed at SuperK with the predicted spectrum is used to build the test statistic to quote the significance to $\bar{\nu}_e$ appearance. The predicted number $N_{SK;r}$ of single e-like ring events in the rth reconstructed energy bin is computed as follows:

$$N_{SK;r} = \sum_m \sum_t \sum_{r\prime} P_{m;t} \cdot T_{r;r\prime;f^{SK}_{E;r}} \cdot S_{m;t;r\prime;\vec{f}} \cdot N^{MC}_{SK;m;r\prime;t}. \quad (1)$$

In Eq. 1, $N^{MC}_{SK;m;r\prime;t}$ is the input SuperK Monte Carlo (MC) template containing the number of events in the single e-like ring MC sample with true reaction mode m in the true energy bin t and the reconstructed energy bin $r\prime$. The construction of the nominal MC templates is discussed in Sec. 2.2. Both flux-tuning weights based on NA61 hadron-production data (see Sec. 2.3) and flux and cross-section tuning weights based on ND280 data (see Sec. 2.4) are applied to the nominal MC templates. $S_{m;t;r\prime;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$ and a vector of nuisance (systematic) parameters $\vec{f}$. A complete description of the nuisance parameters considered in this analysis is given in Sec. 3. $T_{r;r\prime;f^{SK}_{E;r}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the SuperK reconstructed energy scale, expressed here in terms of the nuisance parameter $f^{SK}_{E;r}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the SuperK MC template which corresponds to mode m; application of the term $P_{m;t}$ is discussed in Sec. 2.5.

2.2. Construction of nominal SuperK Monte Carlo templates

The nominal SuperK single e-like ring MC templates $N^{MC}_{SK;m;r;t}$ are constructed by applying the single e-like ring selection cuts to the official SuperK MC samples. Details of the MC samples are given in Sec. 2.2.1. The calculation of the normalization of each MC sample is described in Sec. 2.2.2. The official single e-like ring selection cuts are listed in Sec. 2.2.3. The list of MC templates used in this analysis is shown in Sec. 2.2.4, while the true and reconstructed energy binning of the templates is discussed in Sec. 2.2.5.

2.2.1. Input Monte Carlo samples

This analysis uses the official 14a SuperK MC samples. These were generated using the 13a JNUBEAM flux, NEUT 5.3.2, skdetsim v13p90 and the 14b SuperK software [29]. Six MC samples were generated for antineutrino beam mode running (horn current -250kA). Four of the samples ($\bar{\nu}_\mu$, ν_μ, $\bar{\nu}_e$ and ν_e) were generated using the nominal JNUBEAM flux without oscillations. The fifth sample is $\bar{\nu}_e$ interactions generated with the $\bar{\nu}_\mu$ flux, and will be denoted the "oscillated $\bar{\nu}_e$" sample (it corresponds to the "signal-$\bar{\nu}_e$" sample in the $\bar{\nu}_e$-appearance analysis). The sixth sample is ν_e interactions generated with the ν_μ flux, and will be denoted the "oscillated ν_e" sample.

2.2.2. Normalization of Monte Carlo samples

The normalization (integrated exposure in terms of POT) of each event sample is calculated from the number of events with a MC truth interaction vertex within the 22.5 kt fiducial volume.

$$N = \int dSdIdE \cdot \frac{d^3\Phi_{SK}}{dSdIdE_\nu} \cdot \sigma_{H_2O} \cdot \frac{N_A}{A} \cdot \rho \cdot L \quad (2)$$

where $d^3\Phi_{SK}/dSdIdE$ is the number of flux particles for the given neutrino species per neutrino energy bin dE_ν, per unit area dS and per POT, σ_{H_2O} is the total interaction cross section in water for the given neutrino species, I is the beam intensity in terms of POT, N_A is Avogadro's number, A is the mass number for water, ρ is the water density and L is the neutrino path length in the water volume.

The SuperK MC samples are normalised to the Run 5+6 POT (4.5×10^{20}) using the numbers in Tab. 1. Assuming that all the MC files are used, each sample is first normalised to 10^{21} POT by weighting it with

Sample	Number of events per 22.5 kt fiducial per 10^{21} POT	Number of generated events in true fiducial volume in all available MC
$\bar{\nu}_\mu$	353.789	633988
ν_μ	208.369	634659
$\bar{\nu}_e$	6.811	127013
ν_e	8.941	127262
Oscillated $\bar{\nu}_e$	376.420	126812
Oscillated ν_e	215.099	127278

Table 1: Normalization of input SuperK MC samples [29]

the ratio of the number of events per 22.5 kt per 10^{21} POT (second column) to the number of generated events in the true fiducial volume in all the available MC (third column). Then each sample is normalised to 4.5×10^{20} POT by weighting the result by 0.45(the Run 5+6 POT in units of 10^{21} POT).

2.2.3. Single μ-like ring event selection cuts

The SuperK MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the standard SuperK selection cuts for single e-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts
- the reconstructed vertex is at least 2 metres away from the ID wall
- the visible energy in the event is above 30 MeV
- there is exactly 1 reconstructed ring in the event
- the reconstructed ring is e-like
- number of decay electrons is zero
- the reconstructed neutrino energy is below 1250 MeV
- fiTQun rejection cut: $\ln(L_{\pi^0}/L_e) < 175 - 0.875 \times m_{\gamma\gamma}$

2.2.4. List of Monte Carlo templates

For each SuperK MC sample, a number of different MC templates is constructed corresponding to different true reaction modes. The template granularity depends on the type of oscillation analysis and the specific systematic parameters considered in the analysis. 12 MC templates were used for our 2010 Run 1+2 2-flavour disappearance analysis [2], while 15 MC templates were used for our 2010 Run 1+2 3-flavour disappearance analysis [3], 28 MC templates were used for our 2012 Run 1+2+3 3-flavour disappearance analysis [4] and 32 MC templates were used for our 2014 Run 1+2+3+4 3-flavour disappearance analysis [6].

In our previous ν_μ-disappearance analyses, NC events were divided into NC1$\pi^\pm$ and NC other. For this analysis, like for the $\bar{\nu}_\mu$-disappearance analysis, the NC interaction modes have been further subdivided into NC1$\pi^\pm$, NC1π^0, NC coherent and NC other. We have also added 5 CC MEC templates for $\bar{\nu}_\mu$, ν_μ, $\bar{\nu}_e$, ν_e and oscillated ν_e and 5 templates for the oscillated $\bar{\nu}_e$ MC sample. This gives a total of 46 templates:

- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC MEC,
- $\bar{\nu}_\mu$ CC1π,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC other,

- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC other,
- ν_μ CCQE,
- ν_μ CC MEC,
- ν_μ CC1π,
- ν_μ CC coherent,
- ν_μ CC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1$\pi^\pm$,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1π^0,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC coherent,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC other,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC MEC,
- $\bar{\nu}_e$ CC1π,
- $\bar{\nu}_e$ CC coherent,
- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC other,
- ν_e CCQE,
- ν_e CC MEC,
- ν_e CC1π,
- ν_e CC coherent,
- ν_e CC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1$\pi^\pm$,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1π^0,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC coherent,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC other,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC MEC,

- oscillated $\nu_\mu \to \nu_e$ CC1π,
- oscillated $\nu_\mu \to \nu_e$ CC coherent,
- oscillated $\nu_\mu \to \nu_e$ CC other,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CCQE,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC MEC,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC1π,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC coherent,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC other.

2.2.5. Binning of Monte Carlo templates

Each of the 46 MC templates has 2100 2-dimensional bins (= 84 true energy bins $\times$ 25 reconstructed energy bins). The 84 true energy bins are arranged as follows:

- 6 50-MeV bins from 0-0.3 GeV,
- 28 25-MeV bins from 0.3-1 GeV,
- 40 50-MeV bins from 1-3 GeV,
- 5 100-MeV bin from 3-3.5 GeV,
- 1 bin from 3.5-4 GeV,
- 1 bin from 4-5 GeV,
- 1 bin from 5-7 GeV,
- 1 bin from 7-10 GeV and
- 1 bin from 10-30 GeV.

The 25 reconstructed energy bins are the following:

- 50-MeV bins from 0-1.25 GeV.

The same reconstructed neutrino energy binning is used for the fit of the single e-like ring reconstructed energy spectrum.

A study was performed to decide the optimum binning scheme balancing the needs for accuracy and speed. Details are given in Appendix I of Ref. [4]. The binning described here is unchanged with respect to the Run 1+2+3+4 joint fit analysis,

2.3. Applying flux tuning based on NA61 data

In this analysis, the nominal MC templates, which were constructed using a MC event sample generated with the 13a flux, are reweighted to the 13av1.1 flux which was tuned on NA61 hadron production measurements [31][37]. The reweighted MC templates are denoted the *NA61-tuned* MC templates in this document. The applied weights were obtained from [31] and are shown as a function of the true neutrino energy in Figs. 1-4. The effect of this flux tune on the predicted reconstructed energy spectrum of single e-like ring events is discussed in Sec. 2.6 (see Figs. 1).

	N_{SK} Nominal MC	N_{SK} Flux-tuned MC	N_{SK} BANFF prefit MC	N_{SK} BANFF postfit MC
Total	1.326	1.402	1.247	1.427
ν_μ CCQE	0.012	0.012	0.012	0.014
ν_μ CC 1π	0.005	0.005	0.005	0.005
ν_μ CC coherent	0.000	0.000	0.000	0.000
ν_μ CC MEC	0.002	0.002	0.000	0.002
ν_μ CC other	0.001	0.001	0.001	0.002
ν_μ/ν_τ NC $1\pi^{+/-}$	0.017	0.018	0.018	0.012
ν_μ/ν_τ NC $1\pi^0$	0.068	0.072	0.072	0.045
ν_μ/ν_τ NC coherent	0.021	0.022	0.022	0.023
ν_μ/ν_τ NC other	0.037	0.040	0.040	0.060
$\bar{\nu}_\mu$ CCQE	0.019	0.020	0.018	0.021
$\bar{\nu}_\mu$ CC 1π	0.004	0.004	0.004	0.004
$\bar{\nu}_\mu$ CC coherent	0.001	0.001	0.001	0.002
$\bar{\nu}_\mu$ CC MEC	0.003	0.003	0.001	0.004
$\bar{\nu}_\mu$ CC other	0.000	0.000	0.000	0.000
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	0.013	0.013	0.013	0.009
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	0.086	0.088	0.088	0.054
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	0.082	0.086	0.086	0.095
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	0.050	0.051	0.051	0.082
ν_e CCQE	0.208	0.226	0.212	0.234
ν_e CC 1π	0.054	0.059	0.059	0.046
ν_e CC coherent	0.001	0.001	0.001	0.001
ν_e CC MEC	0.044	0.048	0.013	0.055
ν_e CC other	0.008	0.009	0.009	0.010
ν_e NC $1\pi^{+/-}$	0.001	0.001	0.001	0.000
ν_e NC $1\pi^0$	0.003	0.003	0.003	0.002
ν_e NC coherent	0.001	0.001	0.001	0.001
ν_e NC other	0.002	0.002	0.002	0.003
Osc. ν_e CCQE	0.000	0.000	0.000	0.000
Osc. ν_e CC 1π	0.000	0.000	0.000	0.000
Osc. ν_e CC coherent	0.000	0.000	0.000	0.000
Osc. ν_e CC MEC	0.000	0.000	0.000	0.000
Osc. ν_e CC other	0.000	0.000	0.000	0.000
$\bar{\nu}_e$ CCQE	0.394	0.418	0.371	0.431
$\bar{\nu}_e$ CC 1π	0.079	0.083	0.083	0.076
$\bar{\nu}_e$ CC coherent	0.028	0.029	0.029	0.036
$\bar{\nu}_e$ CC MEC	0.071	0.074	0.020	0.089
$\bar{\nu}_e$ CC other	0.004	0.004	0.004	0.005
$\bar{\nu}_e$ NC $1\pi^{+/-}$	0.000	0.000	0.000	0.000
$\bar{\nu}_e$ NC $1\pi^0$	0.002	0.002	0.002	0.001
$\bar{\nu}_e$ NC coherent	0.001	0.001	0.001	0.001
$\bar{\nu}_e$ NC other	0.001	0.001	0.001	0.002
Osc. $\bar{\nu}_e$ CCQE	0.000	0.000	0.000	0.000
Osc. $\bar{\nu}_e$ CC 1π	0.000	0.000	0.000	0.000
Osc. $\bar{\nu}_e$ CC coherent	0.000	0.000	0.000	0.000
Osc. $\bar{\nu}_e$ CC MEC	0.000	0.000	0.000	0.000
Osc. $\bar{\nu}_e$ CC other	0.000	0.000	0.000	0.000

Table 2: Calculated predicted numbers of 1-ring e-like events without oscillations using the nominal MC templates, the NA61-tuned MC templates, the NIWG-tuned MC templates and the BANFF-tuned MC templates. The BANFF prefit column is the NA61 tuned data, with the CCQE samples reweighted according to the NIWG's external data fits [32]. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 4.5×10^{20}POT.

	N_{SK} Nominal MC	N_{SK} Flux-tuned MC	N_{SK} BANFF prefit MC	N_{SK} BANFF postfit MC
Total	3.890	4.073	3.502	4.183
ν_μ CCQE	0.002	0.002	0.002	0.002
ν_μ CC 1π	0.002	0.002	0.002	0.002
ν_μ CC coherent	0.000	0.000	0.000	0.000
ν_μ CC MEC	0.000	0.000	0.000	0.001
ν_μ CC other	0.001	0.001	0.001	0.001
ν_μ/ν_τ NC $1\pi^{+/-}$	0.017	0.018	0.018	0.012
ν_μ/ν_τ NC $1\pi^0$	0.068	0.072	0.072	0.045
ν_μ/ν_τ NC coherent	0.021	0.022	0.022	0.023
ν_μ/ν_τ NC other	0.037	0.040	0.040	0.060
$\bar{\nu}_\mu$ CCQE	0.003	0.004	0.003	0.004
$\bar{\nu}_\mu$ CC 1π	0.001	0.001	0.001	0.001
$\bar{\nu}_\mu$ CC coherent	0.000	0.000	0.000	0.000
$\bar{\nu}_\mu$ CC MEC	0.000	0.000	0.000	0.000
$\bar{\nu}_\mu$ CC other	0.000	0.000	0.000	0.000
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	0.013	0.013	0.013	0.009
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	0.086	0.088	0.088	0.054
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	0.082	0.086	0.086	0.095
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	0.050	0.051	0.051	0.082
ν_e CCQE	0.191	0.208	0.196	0.215
ν_e CC 1π	0.051	0.055	0.055	0.043
ν_e CC coherent	0.001	0.001	0.001	0.001
ν_e CC MEC	0.041	0.045	0.012	0.051
ν_e CC other	0.008	0.008	0.008	0.009
ν_e NC $1\pi^{+/-}$	0.001	0.001	0.001	0.000
ν_e NC $1\pi^0$	0.003	0.003	0.003	0.002
ν_e NC coherent	0.001	0.001	0.001	0.001
ν_e NC other	0.002	0.002	0.002	0.003
Osc. ν_e CCQE	0.471	0.472	0.446	0.494
Osc. ν_e CC 1π	0.088	0.088	0.088	0.069
Osc. ν_e CC coherent	0.002	0.002	0.002	0.002
Osc. ν_e CC MEC	0.084	0.084	0.023	0.096
Osc. ν_e CC other	0.004	0.004	0.004	0.004
$\bar{\nu}_e$ CCQE	0.367	0.389	0.346	0.401
$\bar{\nu}_e$ CC 1π	0.074	0.078	0.078	0.071
$\bar{\nu}_e$ CC coherent	0.026	0.027	0.027	0.034
$\bar{\nu}_e$ CC MEC	0.066	0.070	0.019	0.083
$\bar{\nu}_e$ CC other	0.004	0.004	0.004	0.005
$\bar{\nu}_e$ NC $1\pi^{+/-}$	0.000	0.000	0.000	0.000
$\bar{\nu}_e$ NC $1\pi^0$	0.002	0.002	0.002	0.001
$\bar{\nu}_e$ NC coherent	0.001	0.001	0.001	0.001
$\bar{\nu}_e$ NC other	0.001	0.001	0.001	0.002
Osc. $\bar{\nu}_e$ CCQE	1.481	1.564	1.395	1.592
Osc. $\bar{\nu}_e$ CC 1π	0.203	0.213	0.213	0.189
Osc. $\bar{\nu}_e$ CC coherent	0.101	0.107	0.107	0.132
Osc. $\bar{\nu}_e$ CC MEC	0.228	0.240	0.065	0.283
Osc. $\bar{\nu}_e$ CC other	0.003	0.003	0.003	0.003

Table 3: Calculated predicted numbers of 1-ring e-like events with oscillations using the nominal MC templates, the NA61-tuned MC templates, the NIWG-tuned MC templates and the BANFF-tuned MC templates. The BANFF prefit column is the NA61 tuned data, with the CCQE samples reweighted according to the NIWG's external data fits [32]. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. The normal hierarchy was assumed, with oscillation parameters give the values listed in table 6. These numbers were calculated for an exposure of 4.5×10^{20}POT.

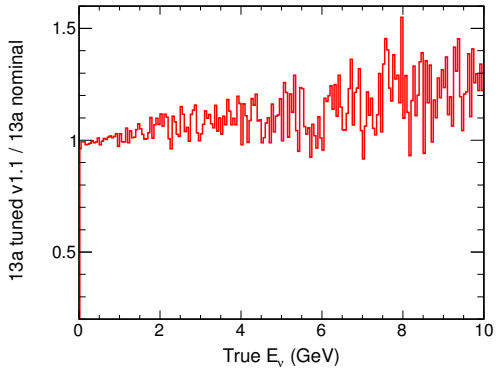


Figure 1: ν_μ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_μ and oscillated-ν_e MC templates [31].

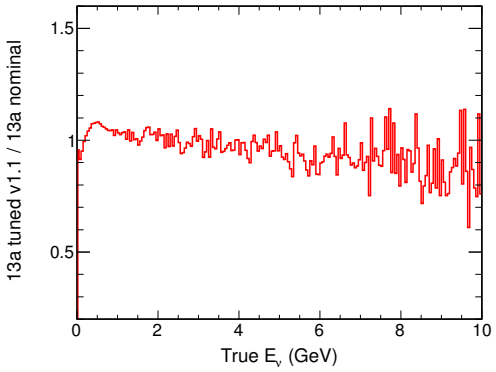


Figure 2: $\bar{\nu}_\mu$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_\mu$ and oscillated-$\bar{\nu}_e$ MC templates [31].

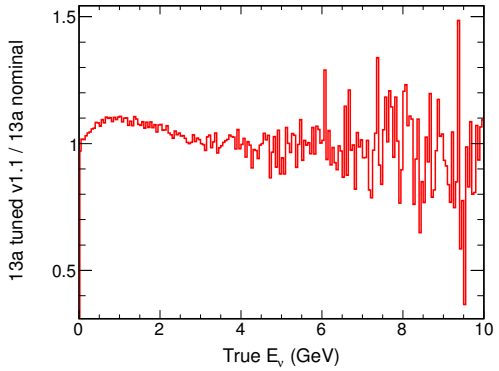


Figure 3: ν_e 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_e MC templates [31].

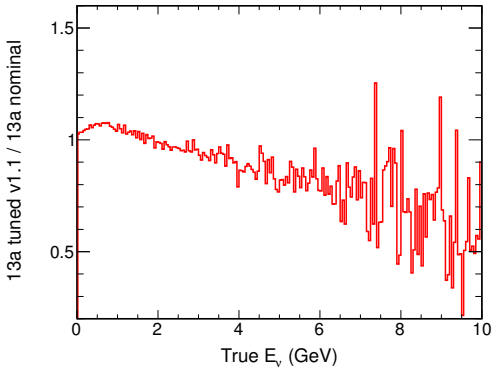


Figure 4: $\bar{\nu}_e$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_e$ MC templates [31].

2.4. Applying flux and cross-section (BANFF) tuning based on ND280 data

The BANFF group has made a fit of the $\bar{\nu}_\mu$ and ν_μ CC0π-like, CC1π-like and CC other ND280 samples to find the best-fit values of the flux normalisation and cross-section parameters[26]. This analysis uses these best-fit values to reweight the NA61-tuned MC templates, and the templates obtained after this second reweighting will be denoted the *BANFF-tuned* MC templates in this document. A list of BANFF parameters, their best-fit values and their 1σ prefit and postfit errors is given in Tab. 4.

The effects of the BANFF tune on the predicted reconstructed energy spectrum of single e-like ring events is discussed in Sec. 2.6 (see Figs. 8 and Tabs. 2 and 3).

2.5. Applying neutrino oscillations

The MC templates are constructed from the unoscillated MC samples and must be weighted with the appropriate survival or oscillation probability. Oscillations are applied as a function of true energy to the MC templates for all CC interactions.

The MC templates calculated from the $\bar{\nu}_\mu$ MC sample are weighted with $\mathrm{P}(\bar{\nu}_\mu \to \bar{\nu}_\mu)$, with this survival probability being calculated using the oscillation parameters, namely $\sin^2\theta_{23}$, $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy), $|\Delta m^2_{21}|$ and δ_{CP}. The $\bar{\nu}_e$ templates are weighted with $\mathrm{P}(\bar{\nu}_e \to \bar{\nu}_e)$, while the oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ templates are weighted with $\mathrm{P}(\bar{\nu}_\mu \to \bar{\nu}_e)$ (the sample was generated assuming 100% of $\bar{\nu}_\mu$ transform to $\bar{\nu}_e$). Again these survival and oscillation probabilities are calculated using the first set of oscillation parameters.

The ν_μ templates are weighted with $\mathrm{P}(\nu_\mu \to \nu_\mu)$. The ν_e templates are weighted with $\mathrm{P}(\nu_e \to \nu_e)$ and the oscillated $\nu_\mu \to \nu_e$ templates with $\mathrm{P}(\nu_\mu \to \nu_e)$, with these survival and oscillation probabilities being calculated using the second set of oscillation parameters.

All these survival and oscillation probabilities are computed in a 3-flavour framework including matter effects in constant-density matter (assuming an Earth crust density of 2.6 g/cm^3). Custom oscillation probability calculation code was developed within the *VALOR* analysis framework. Details and estimates of the numerical accuracy of this code were first presented in [3], and is also shown in Appendix A of [4]. Comparisons between the *VALOR* oscillation probability calculation code and Prob3++ [58], which is typically used by the alternative T2K oscillation analyses, are shown in Appendix B of [4].

In the standard 3-flavour oscillation framework, oscillations of ν_e and ν_μ can yield ν_τ, while oscillations of $\bar{\nu}_e$ and $\bar{\nu}_\mu$ can yield $\bar{\nu}_\tau$. In this analysis, we neglect contributions from ν_τ-CC and $\bar{\nu}_\tau$-CC as their energy threshold is around 3.5 GeV and their effect is negligible[1]. Accordingly, this analysis uses no ν_τ-CC and $\bar{\nu}_\tau$-CC MC templates (and, in fact, no ν_τ and $\bar{\nu}_\tau$ SuperK MC samples are available).

It should be emphasized here that the ν_μ NC MC templates for a mode m are proxies for the NC MC templates for the mixture of ν_e+ν_μ+ν_τ resulting from 3-flavour ν_μ oscillations for that mode m. The same applies to the $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$ NC MC templates. These NC MC templates are unchanged under standard 3-flavour oscillations.

Also it should be noted that there are no explicit NC MC templates made from the oscillated $\bar{\nu}_e$ and ν_e samples. If they were used, the oscillated $\bar{\nu}_e$ (i.e. $\bar{\nu}_e$ coming from oscillations of $\bar{\nu}_\mu$) would be double counted since they are already included in the $\bar{\nu}_\mu$ NC MC templates. The same applies to oscillated ν_e.

2.6. Predictions of nominal and tuned-MC single e-like ring spectra

In this section, we present the expected numbers of events and single e-like ring reconstructed energy spectra in SuperK for various oscillation scenarios. We also present estimates of the effect of the NA61 and BANFF tunings and of assumptions made in the 3-flavour neutrino oscillation framework used in this analysis. Unless explicitly stated otherwise, the oscillation parameters are set to the values shown in Tab. 6 and the normal mass hierarchy is assumed. All plots were generated for an integrated exposure of 4.5×10^{20}POT.

Predicted single e-like ring SuperK reconstructed energy spectra are shown in Fig. 6 for oscillations assuming the normal hierarchy, and the oscillation parameters shown in table 6. For the $\bar{\nu}_e$ appearance analysis the signal is given by the $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ component.

A MC tune derived using NA61 data is applied to the nominal MC templates, and another tune derived using the ND280 data is applied to the NA61-tuned templates as described in Secs. 2.3 and 2.4. The fractional difference in the expected number of single e-like ring events when switching from the nominal MC to the NA61-tuned MC templates is shown in Fig. 7 both oscillated and not oscillated. In Fig. 8 is shown the fractional difference in the expected number of single-ring e-like events when switching from the NA61-tuned MC templates to the BANFF-tuned MC templates. The effects of the NA61 and BANFF tunes on the total expected number of single-ring e-like events and on the numbers in each of the 46 component interaction modes are shown in Tab. 2 for no oscillations, and in Tab. 3 for oscillations assuming the normal hierarchy, and the oscillation parameters shown in table 6. Finally, the effects of the NA61 and BANFF tunes on the reconstructed energy spectrum of single e-like ring events are shown in Figs. 8. In each case, spectra are shown for both no oscillations and for oscillations assuming the normal hierarchy, and the oscillation parameters shown in table 6. Also shown are the ratio of flux-tuned spectrum to nominal spectrum, and BANFF-tuned spectrum to flux-tuned spectrum in Figs. 8.

[1] A back-of-the-envelope calculation can be found in the replies to comments and revision history document of [4] that gives $\ll 1\%$ ν_τ contamination in the 1-ring μ-like SuperK sample. A SuperK MC study is required to determine whether the SuperK efficiency assumptions made there are valid.

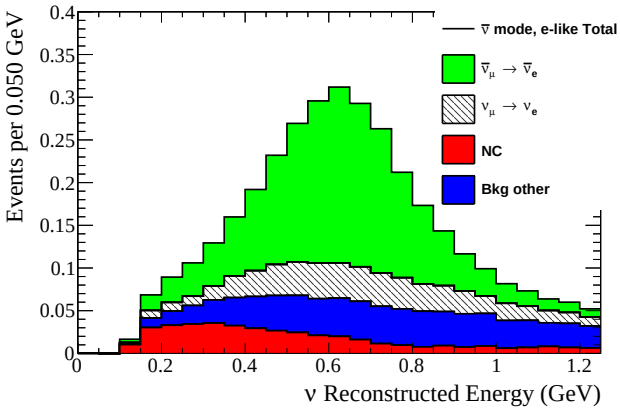


Figure 5: Predicted reconstructed-energy spectrum of 1 e-like ring events, and contributions from various true neutrino reaction modes, with oscillations for an exposure of 4.5×10^{20}POT in antineutrino beam mode. The distribution is broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ signal (green), $\nu_\mu \to \nu_e$ background (grey), NC (red) background and background other (blue) components. The spectrum was generated using the BANFF prefit MC templates. The spectrum shown was generated assuming the normal hierarchy, and the oscillation parameters shown in table 6.

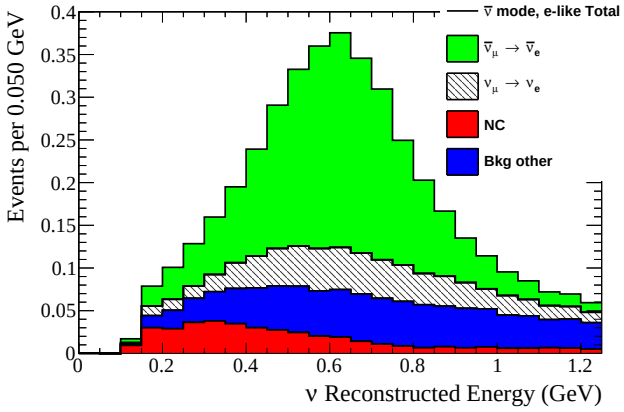


Figure 6: Predicted reconstructed-energy spectrum of 1 e-like ring events, and contributions from various true neutrino reaction modes, with oscillations for an exposure of 4.5×10^{20}POT in antineutrino beam mode. The distribution is broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ signal (green), $\nu_\mu \to \nu_e$ (grey) background, NC background (red) and background other (blue) components. The spectrum was generated using the BANFF postfit-tuned MC templates. The spectrum shown was generated assuming the normal hierarchy, and the oscillation parameters shown in table 6.

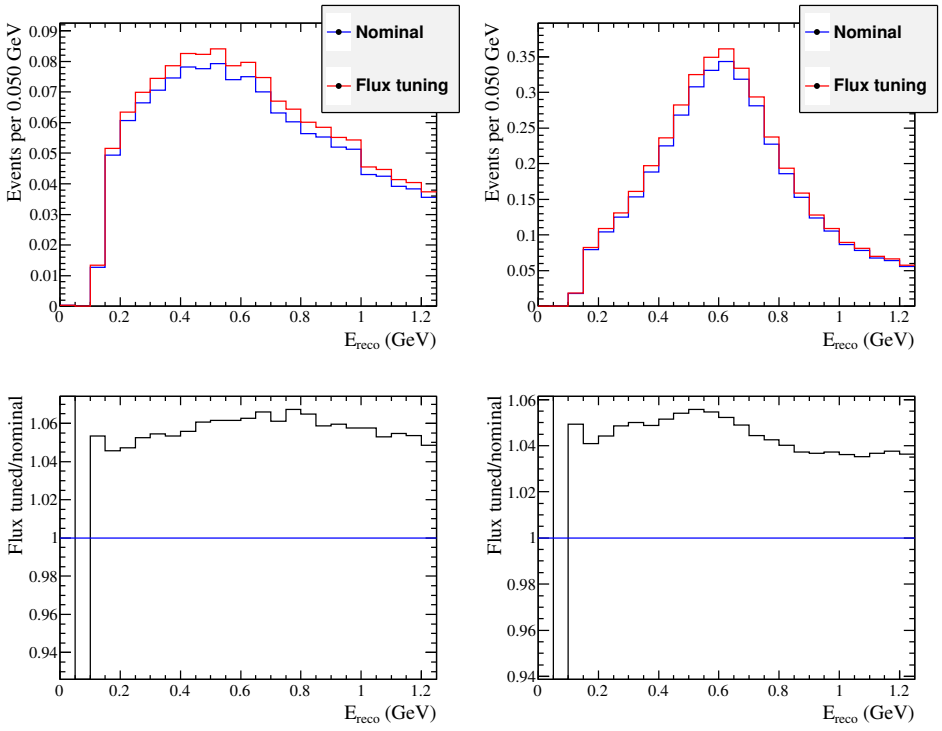


Figure 7: Top: Reconstructed-energy spectrum of 1 e-like ring events, for an exposure of 4.5×10^{20}POT in antineutrino beam mode both with and without the effect of the NA61 flux tuning of the nominal MC templates. The spectra are shown both for no oscillations (left) and for oscillations assuming the normal hierarchy, and the oscillation parameters shown in table 6. Bottom: Ratio of flux-tuned spectrum to the nominal MC spectrum.

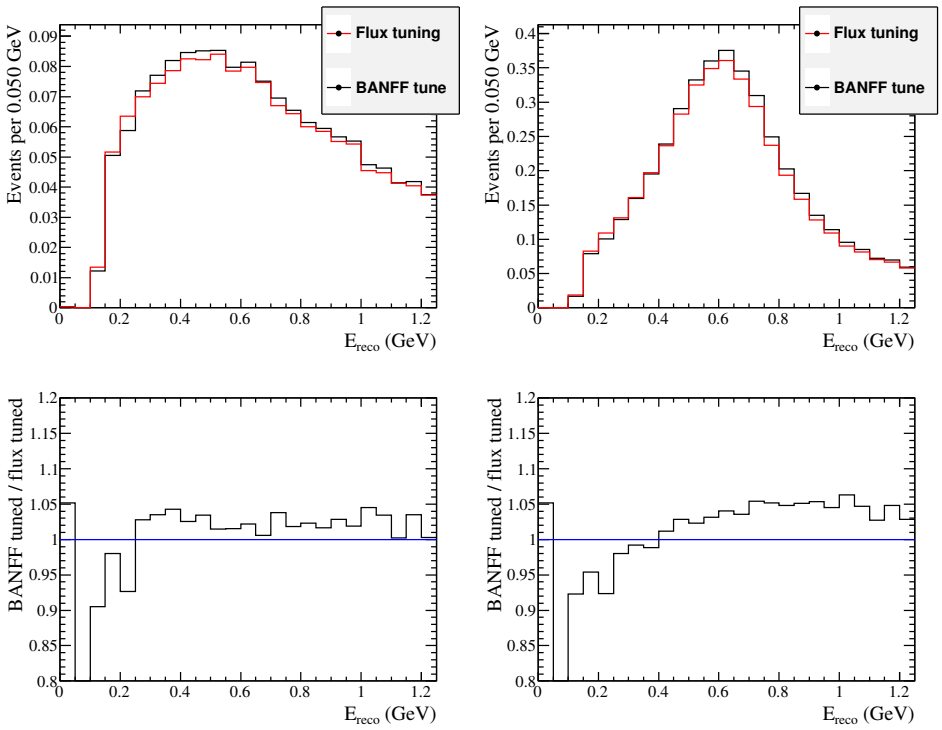


Figure 8: Top: Reconstructed-energy spectrum of 1 e-like ring events, for an exposure of 4.5×10^{20}POT in antineutrino beam mode both with and without the effect of postfit BANFF flux and cross-section tuning on the NA61-tuned flux MC templates. The spectra are shown both for no oscillations (left) and for oscillations assuming the normal hierarchy, and the oscillation parameters shown in table 6. Bottom: Ratio of BANFF-tuned spectrum to the flux-tuned spectrum.

3. Systematic effects in the prediction of the single μ-like ring event energy spectrum

3.1. Input systematic parameters

This analysis considers 56 systematic parameters which can be grouped into 3 categories:

1. **Correlated flux and cross-section (BANFF) parameters** (37 parameters).

In this analysis, all the flux and cross-section systematic parameters are included in the ND280 data (BANFF) fit (in previous ν_μ-disappearance analyses, all the flux parameters but only a subset of the cross-section parameters were included). There are 5 ν_μ, 11 $\bar{\nu}_\mu$, 2 ν_e and 7 $\bar{\nu}_e$ flux parameters as well as 12 cross-section parameters. Best-fit values of these parameters and a covariance matrix giving their correlated uncertainties are obtained by the BANFF group from a fit of the $\bar{\nu}_\mu$ and ν_μ CC0π-like, CC1π-like and CC other-like ND280 samples as described in Ref [26]. A list of the BANFF parameters, their best-fit values and their prefit and postfit errors is given in Tab. 4. The best-fit values are used to reweight the templates as described in Sec. 2.4, and the postfit errors in the covariance matrix are used as prior uncertainties in this analysis.

The effects of these correlated uncertainties are evaluated using the Cholesky method. The covariance matrix is "Cholesky decomposed" into the product of a lower triangular matrix (all elements above the diagonal are zero) and its transpose (in which all elements below the diagonal are zero). Random tweaks of each parameter are independently drawn from a Gaussian distribution with mean zero and standard deviation 1, and these random numbers are put into a vector. This vector is then multiplied by the lower triangular matrix to give a vector of tweaked uncertainties that have the correct correlations between them. The BANFF parameters are then varied by these tweaked uncertainties, with all parameters being varied simultaneously. This process may be repeated as many times as necessary.

Some of the cross-section parameters have a complicated, often non-linear, dependence on true and reconstructed neutrino energy; the effects of uncertainties in these parameters are evaluated by reweighting the MC templates. Weights are calculated for tweaks from -3σ to +3σ in steps of 0.5σ, and interpolations are made between those weights using cubic splines (for MaQE only, weights are calculated from -7σ to +3σ). Cubic splines are made for the following cross-section uncertainties:

- CCQE axial-mass scaling factor (MaQE)
- Fermi momentum for ^{16}O
- Binding energy for ^{16}O
- Nucleon to Δ transition axial form factor (C_5^A)
- Resonance-production axial-mass scaling factor (MaRes)
- Scale of isospin 1/2 nonresonant background
- CC other shape

The splines are made in bins of true and reconstructed energy, and the binning is given in Ref. [30].

The effects of the flux and cross-section normalization uncertainties are also evaluated by reweighting the MC templates, but no splines are made for them since the weight is identical to the tweak for these uncertainties (each flux parameter is a normalization for a limited range of true energy).

The BANFF parameters are applied to the MC templates as specified in Tab. 8.

Parameter	Description	Best fit	1σ pre/postfit fractional error
0	ν_μ flux normalisation, E = 0.0 - 0.7 GeV	1.142	0.094 / 0.071
1	ν_μ flux normalisation, E = 0.7 - 1.0 GeV	1.091	0.079 / 0.053
2	ν_μ flux normalisation, E = 1.0 - 1.5 GeV	1.079	0.077 / 0.052
3	ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.074	0.081 / 0.060
4	ν_μ flux normalisation, E = 2.5 - 30.0 GeV	1.050	0.080 / 0.053
5	$\bar{\nu}_\mu$ flux normalisation, E = 0.0 - 0.4 GeV	1.204	0.104 / 0.069
6	$\bar{\nu}_\mu$ flux normalisation, E = 0.4 - 0.5 GeV	1.210	0.102 / 0.058
7	$\bar{\nu}_\mu$ flux normalisation, E = 0.5 - 0.6 GeV	1.180	0.096 / 0.051
8	$\bar{\nu}_\mu$ flux normalisation, E = 0.6 - 0.7 GeV	1.128	0.085 / 0.041
9	$\bar{\nu}_\mu$ flux normalisation, E = 0.7 - 1.0 GeV	1.123	0.125 / 0.077
10	$\bar{\nu}_\mu$ flux normalisation, E = 1.0 - 1.5 GeV	1.118	0.105 / 0.068
11	$\bar{\nu}_\mu$ flux normalisation, E = 1.5 - 2.5 GeV	1.121	0.080 / 0.053
12	$\bar{\nu}_\mu$ flux normalisation, E = 2.5 - 3.5 GeV	1.121	0.074 / 0.049
13	$\bar{\nu}_\mu$ flux normalisation, E = 3.5 - 5.0 GeV	1.122	0.094 / 0.067
14	$\bar{\nu}_\mu$ flux normalisation, E = 5.0 - 7.0 GeV	1.112	0.093 / 0.064
15	$\bar{\nu}_\mu$ flux normalisation, E = 7.0 - 30.0 GeV	1.093	0.130 / 0.108
16	ν_e flux normalisation, E = 0.0 - 2.5 GeV	1.109	0.069 / 0.047
17	ν_e flux normalisation, E = 2.5 - 30.0 GeV	1.090	0.085 / 0.067
18	$\bar{\nu}_e$ flux normalisation, E = 0.0 - 0.5 GeV	1.193	0.095 / 0.054
19	$\bar{\nu}_e$ flux normalisation, E = 0.5 - 0.7 GeV	1.184	0.091 / 0.047
20	$\bar{\nu}_e$ flux normalisation, E = 0.7 - 0.8 GeV	1.167	0.091 / 0.049
21	$\bar{\nu}_e$ flux normalisation, E = 0.8 - 1.5 GeV	1.141	0.084 / 0.044
22	$\bar{\nu}_e$ flux normalisation, E = 1.5 - 2.5 GeV	1.120	0.080 / 0.054
23	$\bar{\nu}_e$ flux normalisation, E = 2.5 - 4.0 GeV	1.123	0.089 / 0.068
24	$\bar{\nu}_e$ flux normalisation, E = 4.0 - 30.0 GeV	1.180	0.156 / 0.144
25	Meson exchange current normalisation for ^{16}O	1.031	1.041 / 1.015
26	C_5^A nucleon to Δ transition axial form factor	0.854	0.119 / 0.073
27	Scale of isospin 1/2 nonresonant background	1.143	0.154 / 0.146
28	CCQE axial-mass scaling factor	0.948	0.058 / 0.028
29	Resonance-production axial-mass scaling factor	0.762	0.158 / 0.055
30	Fermi momentum for ^{16}O	0.997	0.055 / 0.054
31	CC other shape	0.023	0.400 / 0.193
32	Binding energy for ^{16}O	1.002	0.333 / 0.326
33	CC coherent for ^{16}O normalisation	1.076	1.000 / 0.972
34	NC coherent normalisation	0.980	0.300 / 0.299
35	NC other normalisation	1.413	0.300 / 0.186
36	CC ν_e normalisation	1.001	0.030 / 0.030

Table 4: Summary of flux and correlated cross-section (BANFF) systematics included in the *VALOR* Run 5+6 $\bar{\nu}_e$-appearance analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [26].

Parameter	Definition	1σ error
1	oscillated ν_e CC, $E_{reco} < 0.35$ GeV	0.059
2	oscillated ν_e CC, $0.35 < E_{reco} < 0.80$ GeV	0.027
3	oscillated ν_e CC, $0.80 < E_{reco} < 1.25$ GeV	0.056
4	ν_μ and $\bar{\nu}_\mu$ CC, $E_{reco} < 0.35$ GeV	1.241
5	ν_μ and $\bar{\nu}_\mu$ CC, $0.35 < E_{reco} < 0.80$ GeV	1.129
6	ν_μ and $\bar{\nu}_\mu$ CC, $0.80 < E_{reco} < 1.25$ GeV	0.917
7	ν_e and $\bar{\nu}_e$ CC, $E_{reco} < 0.35$ GeV	0.044
8	ν_e and $\bar{\nu}_e$ CC, $0.35 < E_{reco} < 0.80$ GeV	0.032
9	ν_e and $\bar{\nu}_e$ CC, $0.80 < E_{reco} < 1.25$ GeV	0.068
10	all NC, $E_{reco} < 0.35$ GeV	0.476
11	all NC, $0.35 < E_{reco} < 0.80$ GeV	0.301
12	all NC, $0.80 < E_{reco} < 1.25$ GeV	0.608

Table 5: Binning of SuperK selection efficiency and FSI+SI systematic parameters in covariance matrices

2. **SuperK efficiencies & effects of intranuclear and secondary re-interactions** (12 parameters).

There are uncertainties in the efficiencies of the fiducial volume and reduction chain, and the OD, ring-counting, PID, momentum and decay-electron cuts. These uncertainties have been evaluated by comparing 1412 days of SK-IV atmospheric data with atmospheric neutrino Monte Carlo. Ring counting errors are evaluated using ν_μ CCQE, ν_μ CCnQE, NC, and ν_e CC-enriched control samples. The NC PID error is evaluated using a NC-enriched control sample, which does not use PID information. Each Monte Carlo control sample is fitted to the data using the efficiency as a fit parameter, and the uncertainties in the efficiencies are estimated by adding the covariance matrix containing the differences between the best-fit and nominal efficiencies to the covariance matrix containing the statistical errors in the fitted efficiencies [27].

Intranuclear final-state interactions (FSI) have significant effects in the T2K energy range and result in event topologies, for scattering from nuclear targets, which are drastically different from those created in neutrino scattering from free nucleons. Uncertainties due to FSI were estimated by simultaneously varying the NEUT parameters that scale the interaction probabilities for quasi-elastic scattering, absorption, charge exchange and pion production using the method described in Sec. 2 in [18]. The NIWG group produced 16 parameter sets that span the 1σ surface in the parameter space; this was done by varying the NEUT parameters when running the pion scattering simulations, and comparing the results with external pion-carbon scattering data. These 16 parameter sets are shown in Tab. 2 in [18]. Uncertainties due to secondary pion interactions (SI) were also evaluated. The NIWG group adapted the NEUT nuclear cascade model to simulate SI in the SuperK detector simulation (SKDETSIM) and to allow variations in the interaction probabilities. Details of this are given in [21]. Since the same model is used for FSI and SI, it is possible to evaluate the uncertainties in both FSI and SI simultaneously. This was done by reweighting a single e-like ring selected SK MC with the 16 FSI parameter sets using T2KReweight. The reweighting method for FSI is described in Sec. 8 in [17]. Variations in the SI interaction probabilities were included for each FSI parameter set. Subsequently a covariance matrix was calculated using equation 1 in [18]; the quantity p in that equation is the average event weight for a FSI parameter set, and the sum is over the parameter sets.

We consider the SuperK detector efficiency and FSI+SI systematics together, and add their covariance matrices linearly. Each of these covariance matrices has 12 correlated parameters, corresponding

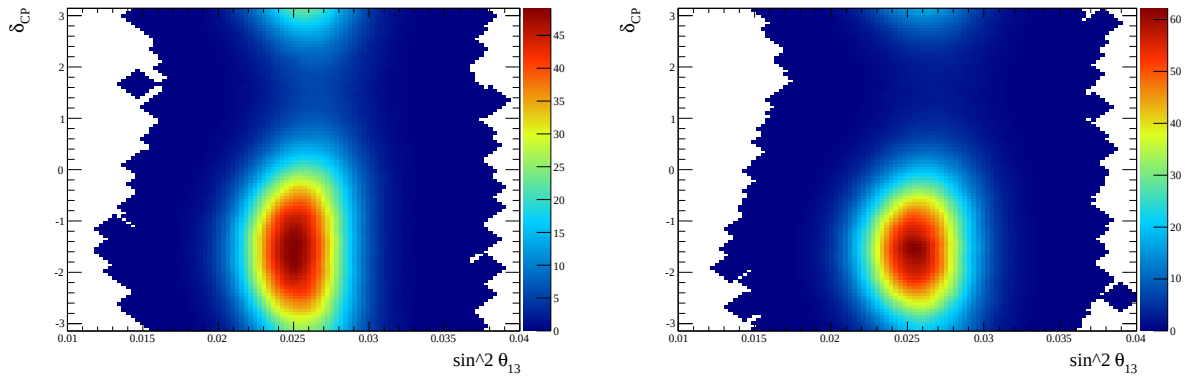


Figure 9: Prior distributions used to randomly extract $\sin^2\theta_{13}$ and δ_{CP}. The correlations between the parameters are taken into account. The distributions are from the T2K+reactor joint Mach3 bayesian analysis [36].

Parameter(s)	Treatment	Nominal value
$\sin^2\theta_{23}$	marginalized	0.528
$\sin^2\theta_{13}$	marginalized	0.025
$\sin^2\theta_{12}$	fixed	0.306
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	marginalized	2.509×10^{-3} eV^2/c^4
Δm^2_{21}	fixed	7.5×10^{-5} eV^2/c^4
δ_{CP}	marginalized	-1.601
Mass Hierarchy	marginalized	NH

Table 6: Treatment of the oscillation parameters in the p-value computation as well as used values for fixed parameters. The values of the fixed oscillation parameters correspond to the most probable parameters of the bayesian analysis described in [13, 36]. The mass hierarchy is marginalized as well.

to different modes and reconstructed energy bins, as shown in Tab. 5. The effects of these uncertainties are also evaluated using the Cholesky method.

3. **SuperK energy scale** (1 parameter).

The systematic parameter $f^{SK}_{E;r}$ is included to estimate the effects of uncertainty in the SuperK reconstructed energy scale. This uncertainty is estimated to be 2.4% [28]; its effects are calculated by scaling the bin edges of the MC templates and, assuming uniform distribution of events within the bins, calculating the number of events gained from / lost to neighbouring bins. The effect of the energy scale uncertainty is propagated to all MC templates as shown in Tab. 8.

4. **Oscillation parameters and mass hierarchy** (5 parameters).

The 3-flavour oscillation probability is used. The mass hierarchy and the following oscillation parameters are randomly extracted from the posterior distribution of the T2K+reactor joint Mach3 bayesian analysis [36]. A discrete prior is used for the mass hierarchy: the probability for normal hierarchy (NH) and inverted hierarchy (IH) is respectively 0.684 and 0.326. Different priors of the oscillation parameters are used for NH and IH. $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for NH) ($|\Delta m^2_{31}|$ for IH) are randomly extracted from the 2D prior distribution shown in Fig. 10, taking into account the existing correlations. $\sin^2\theta_{13}$ and δ_{CP} are extracted from the 2D prior distribution shown in Fig. 9 taking into account the correlations.

The treatment of the oscillation parameters in the analysis as well as the nominal values are shown in Tab. 6.

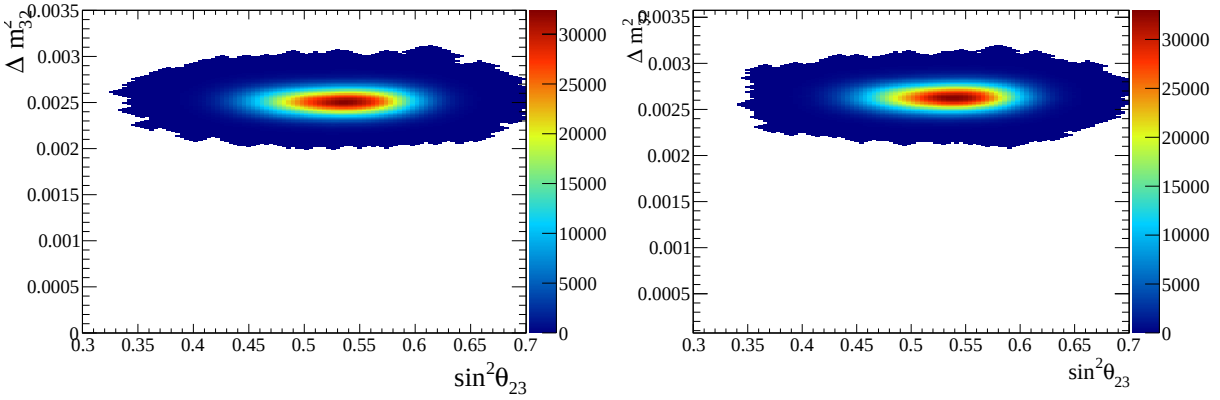


Figure 10: Prior distributions used to randomly extract $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for NH (left), $\sin^2\theta_{23}$ and $|\Delta m^2_{31}|$ for IH (right). The correlations between the parameters are taken into account. The distributions are from the T2K+reactor joint Mach3 bayesian analysis [36].

Parameter(s)	1σ effect (Std. Dev. / Mean)
$\sin^2\theta_{12}$	-
Δm^2_{21}	-
$\sin^2\theta_{23}$	0.080
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	0.042
$\sin^2\theta_{13}$	0.080
δ_{CP}	0.084
$\sin^2\theta_{13}$ & δ_{CP}	0.114
$\sin^2\theta_{23}$ & $\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	0.080
all	0.130

Table 7: 1σ effect on the total number of superK events as the standard deviation divided by the mean of the distribution given by the oscillation parameters. $\sin^2\theta_{12}$ and Δm^2_{21} are kept fixed. The mass hierarchy is always randomly extracted. The priors described above are used and correlations between parameters are taken into account when existing. 10k toys are performed.

The effect of the oscillation parameters on the total number of superK events is shown in Tab. 7. How the effect of each single oscillation parameter is mixed with the others is not trivial. Indeed priors with correlations for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ (NH) / $|\Delta m^2_{31}|$ (IH) and for $\sin^2\theta_{13}$ and δ_{CP} are used. Furthermore, since a 3-flavour oscillation probability is used, when the oscillation parameters are randomly extracted together additional correlations are introduced as well.

	$f^{SK}_{E;r}$	$f^{SK+FSI}_{0;t,r}$ RHC	$f^{SK+FSI}_{1;t,r}$ RHC	$f^{SK+FSI}_{2;t,r}$ RHC	$f^{SK+FSI}_{3;t,r}$ RHC	$f^{SK+FSI}_{4;t,r}$ RHC	$f^{SK+FSI}_{5;t,r}$ RHC	$f^{SK+FSI}_{6;t,r}$ RHC	$f^{SK+FSI}_{7;t,r}$ RHC	$f^{SK+FSI}_{8;t,r}$ RHC	$f^{SK+FSI}_{9;t,r}$ RHC	$f^{SK+FSI}_{10;t,r}$ RHC	$f^{SK+FSI}_{11;t,r}$ RHC	$f^{banff}_{0;t,r}$ RHC	$f^{banff}_{1;t,r}$ RHC	$f^{banff}_{2;t,r}$ RHC	$f^{banff}_{3;t,r}$ RHC	$f^{banff}_{4;t,r}$ RHC	$f^{banff}_{5;t,r}$ RHC	$f^{banff}_{6;t,r}$ RHC	$f^{banff}_{7;t,r}$ RHC
ν_μ CCQE	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC 1π	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC coherent	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC MEC	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC other	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC $1\pi^{+/-}$	✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC $1\pi^0$	✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC coherent	✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC other	✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu$ CCQE	✓				✓	✓	✓														
$\bar{\nu}_\mu$ CC 1π	✓				✓	✓	✓														
$\bar{\nu}_\mu$ CC coherent	✓				✓	✓	✓														
$\bar{\nu}_\mu$ CC MEC	✓				✓	✓	✓														
$\bar{\nu}_\mu$ CC other	✓				✓	✓	✓														
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	✓										✓	✓	✓								
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	✓										✓	✓	✓								
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	✓										✓	✓	✓								
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	✓										✓	✓	✓								
ν_e CCQE	✓							✓	✓	✓											
ν_e CC 1π	✓							✓	✓	✓											
ν_e CC coherent	✓							✓	✓	✓											
ν_e CC MEC	✓							✓	✓	✓											
ν_e CC other	✓							✓	✓	✓											
ν_e NC $1\pi^{+/-}$	✓										✓	✓	✓								
ν_e NC $1\pi^0$	✓										✓	✓	✓								
ν_e NC coherent	✓										✓	✓	✓								
ν_e NC other	✓										✓	✓	✓								
Osc. ν_e CCQE	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC 1π	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC coherent	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC MEC	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC other	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_e$ CCQE	✓							✓	✓	✓											
$\bar{\nu}_e$ CC 1π	✓							✓	✓	✓											
$\bar{\nu}_e$ CC coherent	✓							✓	✓	✓											
$\bar{\nu}_e$ CC MEC	✓							✓	✓	✓											
$\bar{\nu}_e$ CC other	✓							✓	✓	✓											
$\bar{\nu}_e$ NC $1\pi^{+/-}$	✓										✓	✓	✓								
$\bar{\nu}_e$ NC $1\pi^0$	✓										✓	✓	✓								
$\bar{\nu}_e$ NC coherent	✓										✓	✓	✓								
$\bar{\nu}_e$ NC other	✓										✓	✓	✓								
Osc. $\bar{\nu}_e$ CCQE	✓	✓	✓	✓																	
Osc. $\bar{\nu}_e$ CC 1π	✓	✓	✓	✓																	
Osc. $\bar{\nu}_e$ CC coherent	✓	✓	✓	✓																	
Osc. $\bar{\nu}_e$ CC MEC	✓	✓	✓	✓																	
Osc. $\bar{\nu}_e$ CC other	✓	✓	✓	✓																	

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Table 8 – *Continued from previous page*

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3.2. Evaluation of effects of systematic parameters on the predicted single e-like ring reconstructed energy spectrum

In this section, we present the effects of the systematics described in Sec. 3.1 on the predicted number of single e-like ring events and the predicted reconstructed energy spectrum in SuperK. These effects are shown both without oscillations and with oscillations with the values shown in Tab. 6. All results are generated for an integrated exposure of $4.5\times10^{20}$and the BANFF-tuned MC templates are used.

The effects of systematic uncertainties on the predicted number of events are summarised in Tab. 10, both for no oscillations and for oscillations with the typical parameter values given above. In this table, systematic parameters are grouped into five categories and all parameter correlations are taken into account. Percentage uncertainties are given for both the BANFF prefit and postfit errors.

The effects of the combined systematic uncertainties on the reconstructed energy spectrum of single e-like ring events is shown in Fig. 12, 13, 14 and 15 for null oscillations and Fig. 16, 17, 18 and 19 for the typical oscillation scenario. This plot shows the total error envelope for the 1 e-like ring event reconstructed energy spectrum; this is calculated as the $\pm1\sigma$ spread of bin contents from 10k toy MC experiments generated with randomized systematic parameters. All systematic parameters were considered and their correlations were taken into account.

The largest uncertainty is the SuperK detector efficiency/FSI parameter for NC events. The uncertainty in this parameter has a large effect on the oscillation parameters since the uncertainty itself is large (please see Tab. 5) and NC events constitute a significant fraction of the number of events near the oscillation maximum (please see figure 6). The second-largest uncertainty is the MEC normalization parameter; this has a large effect on the oscillation parameters since the uncertainty itself is large at 102% (please see table 4); the reason for the size of this uncertainty is that the rate of MEC events on oxygen is not constrained by the BANFF fit of ND280 data.

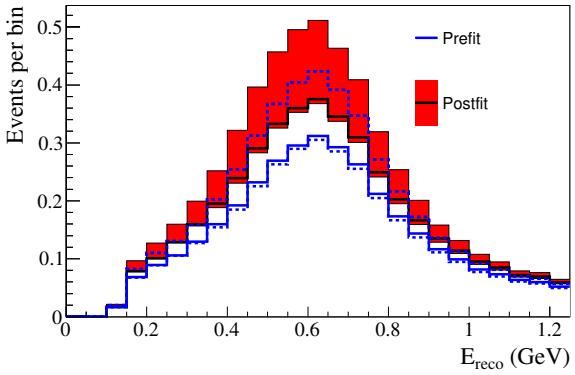


Figure 11: Error envelopes for the energy spectrum as a result of oscillation parameters. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
All syst	3.88805	4.1829	5.805	38.779 %	2.89312	-30.835 %
$f^{banff}_{0;t,r}RHC$	3.88805	4.18289	4.2021	0.45919 %	4.16368	-0.45919 %
$f^{banff}_{1;t,r}RHC$	3.88805	4.18289	4.19574	0.30729 %	4.17004	-0.30729 %
$f^{banff}_{2;t,r}RHC$	3.88805	4.18289	4.18942	0.15616 %	4.17636	-0.15616 %
$f^{banff}_{3;t,r}RHC$	3.88805	4.18289	4.18544	0.060979 %	4.18034	-0.060979 %
$f^{banff}_{4;t,r}RHC$	3.88805	4.18289	4.1857	0.067114 %	4.18008	-0.067114 %
$f^{banff}_{5;t,r}RHC$	3.88805	4.18289	4.18569	0.067044 %	4.18009	-0.067044 %
$f^{banff}_{6;t,r}RHC$	3.88805	4.18289	4.19237	0.22667 %	4.17341	-0.22667 %
$f^{banff}_{7;t,r}RHC$	3.88805	4.18289	4.20526	0.53488 %	4.16052	-0.53488 %
$f^{banff}_{8;t,r}RHC$	3.88805	4.18289	4.2069	0.57395 %	4.15888	-0.57395 %
$f^{banff}_{9;t,r}RHC$	3.88805	4.18289	4.2377	1.3103 %	4.12808	-1.3103 %
$f^{banff}_{10;t,r}RHC$	3.88805	4.18289	4.19164	0.20919 %	4.17414	-0.20919 %
$f^{banff}_{11;t,r}RHC$	3.88805	4.18289	4.18441	0.036372 %	4.18137	-0.036372 %
$f^{banff}_{12;t,r}RHC$	3.88805	4.18289	4.18353	0.015292 %	4.18225	-0.015292 %
$f^{banff}_{13;t,r}RHC$	3.88805	4.18289	4.18373	0.020058 %	4.18205	-0.020058 %
$f^{banff}_{14;t,r}RHC$	3.88805	4.18289	4.18316	0.0063628 %	4.18262	-0.0063628 %
$f^{banff}_{15;t,r}RHC$	3.88805	4.18289	4.18317	0.0067478 %	4.18261	-0.0067478 %
$f^{banff}_{16;t,r}RHC$	3.88805	4.18289	4.19665	0.32891 %	4.16913	-0.32891 %
$f^{banff}_{17;t,r}RHC$	3.88805	4.18289	4.18308	0.0045157 %	4.1827	-0.0045157 %
$f^{banff}_{18;t,r}RHC$	3.88805	4.18289	4.18792	0.12033 %	4.17786	-0.12033 %
$f^{banff}_{19;t,r}RHC$	3.88805	4.18289	4.18835	0.13055 %	4.17743	-0.13055 %
$f^{banff}_{20;t,r}RHC$	3.88805	4.18289	4.1859	0.071987 %	4.17988	-0.071987 %
$f^{banff}_{21;t,r}RHC$	3.88805	4.18289	4.19309	0.24384 %	4.17269	-0.24384 %
$f^{banff}_{22;t,r}RHC$	3.88805	4.18289	4.18364	0.01785 %	4.18214	-0.01785 %
$f^{banff}_{23;t,r}RHC$	3.88805	4.18289	4.18297	0.001949 %	4.18281	-0.001949 %
$f^{banff}_{24;t,r}RHC$	3.88805	4.18289	4.18297	0.0019185 %	4.18281	-0.0019185 %
$f^{banff}_{Norm_{MEC}}RHC$	3.88805	4.18289	4.68954	12.112 %	3.67624	-12.112 %
$f^{banff}_{C^A_5}RHC$	3.88805	4.18289	4.23627	1.2761 %	4.13791	-1.0754 %
$f^{banff}_{Bg_{RES}}RHC$	3.88805	4.18289	4.20599	0.55227 %	4.16257	-0.48583 %
$f^{banff}_{M^{QE}_A}RHC$	3.88805	4.18289	4.23291	1.1958 %	4.13268	-1.2004 %
$f^{banff}_{M^{RES}_A}RHC$	3.88805	4.18289	4.20323	0.48629 %	4.16495	-0.42893 %
$f^{banff}_{SCCA}RHC$	3.88805	4.18289	4.18289	0 %	4.18289	0 %
$f^{banff}_{SCCV}RHC$	3.88805	4.18289	4.18289	0 %	4.18289	0 %
$f^{banff}_{p_f}RHC$	3.88805	4.18289	4.12908	-1.2865 %	4.23618	1.2739 %
$f^{banff}_{Shape_{CCoth}}RHC$	3.88805	4.18289	4.18598	0.073756 %	4.17981	-0.073755 %
$f^{banff}_{E_B}RHC$	3.88805	4.18289	4.18356	0.015919 %	4.18226	-0.015148 %
$f^{banff}_{Norm_{CCcoh}}RHC$	3.88805	4.18289	4.3362	3.6653 %	4.02958	-3.6653 %
$f^{banff}_{Norm_{NCcoh}}RHC$	3.88805	4.18289	4.21974	0.88103 %	4.14604	-0.88103 %
$f^{banff}_{Norm_{NCother}}RHC$	3.88805	4.18289	4.20228	0.46352 %	4.1635	-0.46352 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}RHC$	3.88805	4.18289	4.29622	2.7093 %	4.06956	-2.7093 %
$f^{SK}_{E;r}RHC$	3.88805	4.18289	4.18289	0 %	4.18289	2.1234e-14 %
$f^{SK+FSI}_{0;t,r}RHC$	3.88805	4.18289	4.19724	0.34304 %	4.16854	-0.34304 %
$f^{SK+FSI}_{1;t,r}RHC$	3.88805	4.18289	4.23821	1.3225 %	4.12757	-1.3225 %
$f^{SK+FSI}_{2;t,r}RHC$	3.88805	4.18289	4.21541	0.77741 %	4.15037	-0.77741 %
$f^{SK+FSI}_{3;t,r}RHC$	3.88805	4.18289	4.18901	0.14636 %	4.17677	-0.14636 %
$f^{SK+FSI}_{4;t,r}RHC$	3.88805	4.18289	4.18688	0.095311 %	4.1789	-0.095311 %
$f^{SK+FSI}_{5;t,r}RHC$	3.88805	4.18289	4.18535	0.058848 %	4.18043	-0.058848 %
$f^{SK+FSI}_{6;t,r}RHC$	3.88805	4.18289	4.18707	0.099886 %	4.17871	-0.099886 %
$f^{SK+FSI}_{7;t,r}RHC$	3.88805	4.18289	4.19758	0.35113 %	4.1682	-0.35113 %
$f^{SK+FSI}_{8;t,r}RHC$	3.88805	4.18289	4.20729	0.58334 %	4.15849	-0.58334 %
$f^{SK+FSI}_{9;t,r}RHC$	3.88805	4.18289	4.2507	1.6212 %	4.11508	-1.6212 %
$f^{SK+FSI}_{10;t,r}RHC$	3.88805	4.18289	4.24007	1.367 %	4.12571	-1.367 %
$f^{SK+FSI}_{11;t,r}RHC$	3.88805	4.18289	4.21843	0.8496 %	4.14735	-0.8496 %

Table 9: $\pm1\sigma$ effect of the systematic parameters on the total number of superK events.

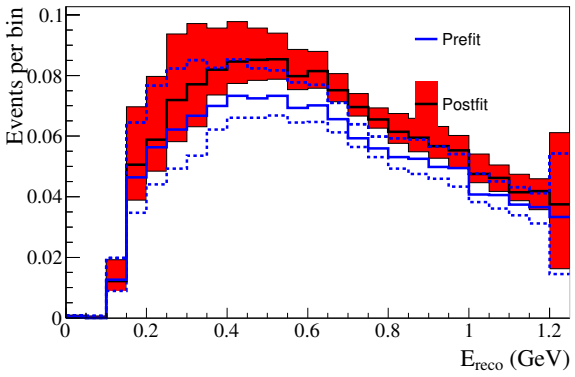


Figure 12: Error envelopes for the energy spectrum as a result of SuperK detector and FSI/SI parameters. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. No oscillations were applied.

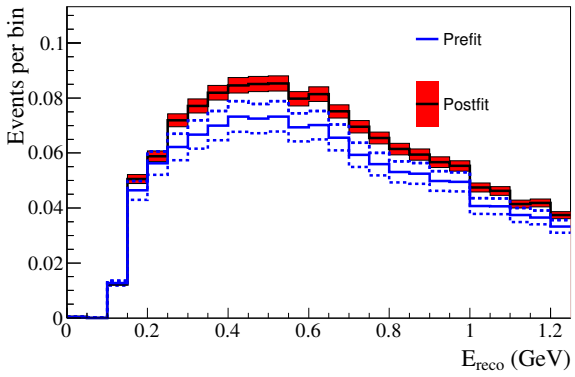


Figure 13: Error envelopes for the energy spectrum as a result of BANFF flux systematics. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. No oscillations were applied.

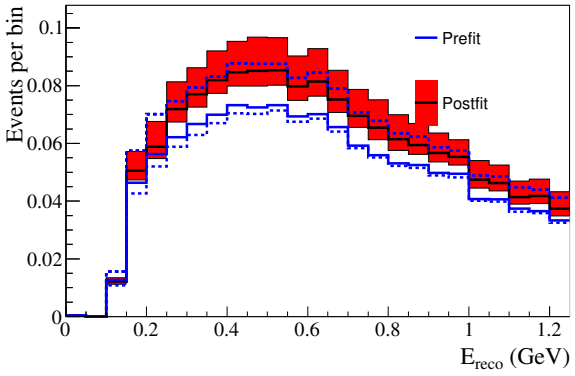


Figure 14: Error envelopes for the energy spectrum as a result of only cross-section BANFF parameters. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. No oscillations were applied.

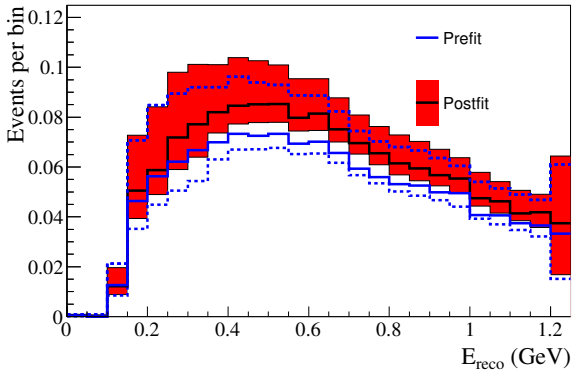


Figure 15: Error envelopes for the energy spectrum as a result of all systematic parameters. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. No oscillations were applied.

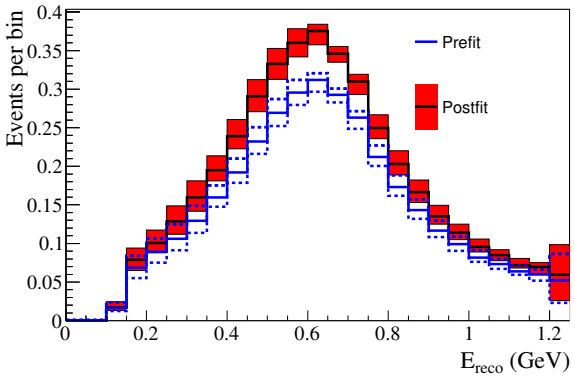


Figure 16: Error envelopes for the energy spectrum as a result of SuperK detector and FSI/SI parameters. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were fixed to the parameter values shown in table 6.

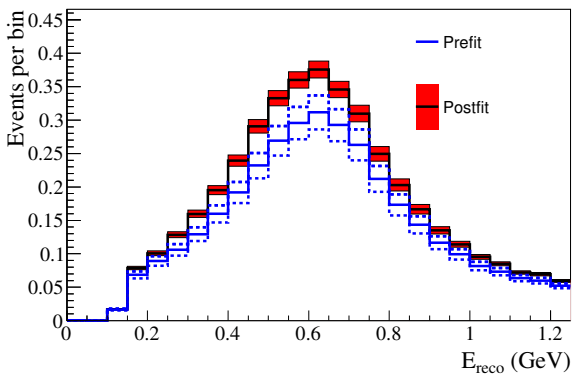


Figure 17: Error envelopes for the energy spectrum as a result of BANFF flux systematics. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were fixed to the parameter values shown in table 6.

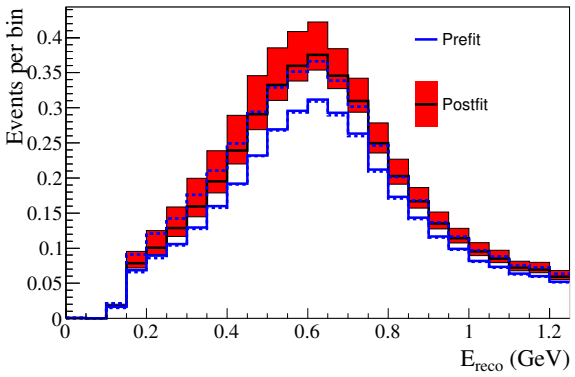


Figure 18: Error envelopes for the energy spectrum as a result of only cross-section BANFF parameters. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were fixed to the parameter values shown in table 6.

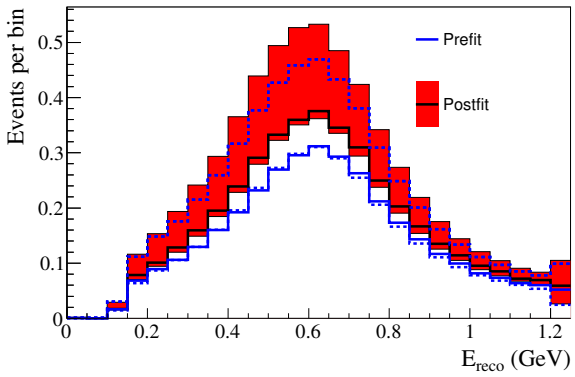


Figure 19: Error envelopes for the energy spectrum as a result of all systematic parameters. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were treated as shown in table 6: $\sin^2\theta_{23}$, $\sin^2\theta_{13}$, Δm^2_{23} and δ_{CP} were marginalized.

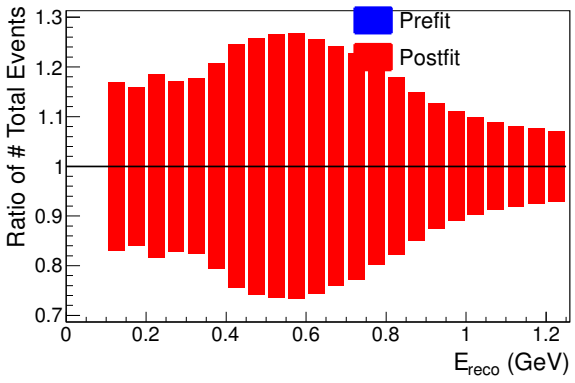


Figure 20: Error envelopes, expressed a ratio of event count, for the energy spectrum as a result of oscillation parameters. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were applied with the parameter values shown in table 6.

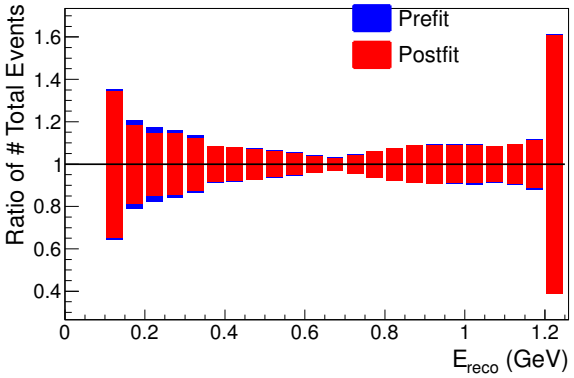


Figure 21: Error envelopes, expressed a ratio of event count, for the energy spectrum as a result of SuperK detector and FSI/SI parameters. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were applied with the parameter values shown in table 6.

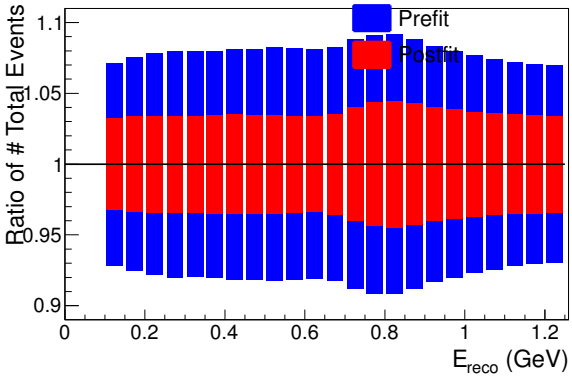


Figure 22: Error envelopes, expressed a ratio of event count, for the energy spectrum as a result of BANFF flux systematics. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were applied with the parameter values shown in table 6.

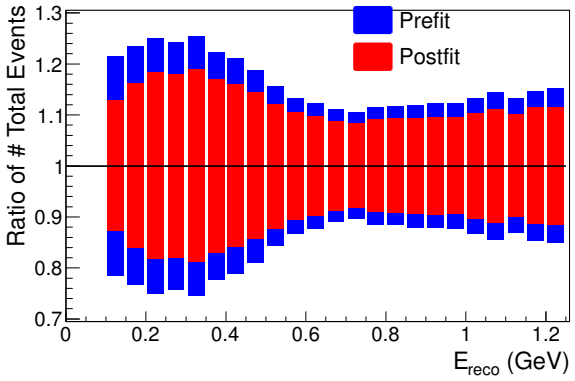


Figure 23: Error envelopes, expressed a ratio of event count, for the energy spectrum as a result of only cross-section BANFF parameters. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were applied with the parameter values shown in table 6.

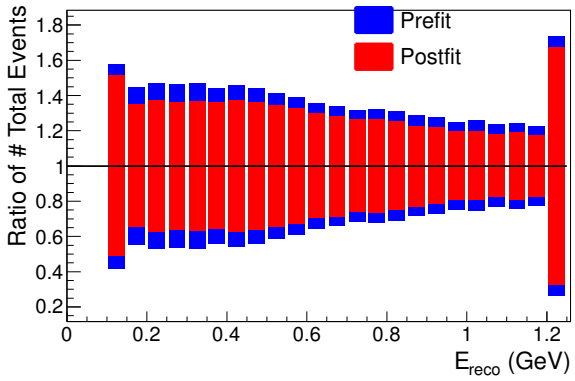


Figure 24: Error envelopes, expressed a ratio of event count, for the energy spectrum as a result of all systematic parameters. 10k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were applied with the parameter values shown in table 6.

Source of uncertainty	$\delta N_{SK}/< N_{SK} >$ (no oscillations)	$\delta N_{SK}/< N_{SK} >$ (oscillations with typical parameter values)
SuperK detector + FSI + SI	9.15%	3.88%
BANFF (prefit)	12.43%	12.75%
BANFF (postfit)	9.21%	10.59%
BANFF (Non-oxygen, postfit)	3.96%	3.05%
BANFF (Oxygen-specific postfit)	8.47%	10.22%
Oscillation parameters	0.00%	13.43%
Total (BANFF prefit)	15.44%	18.92%
Total (BANFF postfit)	12.98%	17.54%

Table 10: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events in the reverse-horn current sample. The BANFF parameters considered to be oxygen-specific are Oxygen MEC normalisation, Fermi Energy, Binding Energy, CC Coherent Normalisation, NC Coherent normalization and the nue/numu normalisation. The effect of each group of systematics is computed as the standard deviation divided by the mean of the distribution of the total number of superK events.

4. Analysis procedure

4.1. Fit method

The objective of this analysis is to search for $\bar{\nu}_e$ appearance from a $\bar{\nu}_\mu$ beam. The parameter of interest is β, a not physical parameter which was introduced to normalize the $\bar{\nu}_e$ appearance signal component (please see figure 6). The hypothesis of null $\bar{\nu}_e$ appearance is obtained setting $\beta = 0$, while $\bar{\nu}_e$ appearance corresponding to the current best knowledge of the PMNS matrix is given by $\beta = 1$. All the systematic parameters are marginalized. Only the solar parameters $\sin^2\theta_{12}$ and Δm^2_{21} are fixed because they have a negligible impact on the oscillations at the T2K energy. The other oscillation parameters are marginalized using the prior distributions which correspond to the posterior of the T2K+reactor joint fit bayesian analysis, which are shown in section 3.1. The treatment of the oscillation parameters is shown in table 6.

The *VALOR* appearance analysis uses a binned likelihood-ratio method where the estimation of the β parameter is obtained by comparing the observed and predicted SuperK reconstructed energy spectra for 1-ring e-like events.

Let N be the number of reconstructed energy bins and $\mathbf{a}$ be a $1 \times N_s$ - dimensional array of systematic (nuisance) parameters that affect the SuperK reconstructed energy spectrum prediction (e.g parameters that control the beam, neutrino interaction simulations, detector response and oscillations). The log-likelihood function used in the analysis is

$$-2\ln\lambda(\beta,\mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot\ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp}-n_i^{obs})\right) + (\mathbf{a}-\mathbf{a_0})^T\cdot\mathbf{C}^{-1}\cdot(\mathbf{a}-\mathbf{a_0}) \tag{3}$$

where $\mathbf{a_0}$ is an $1 \times N_s$ - dimensional array with the default values of the systematics parameters, $\mathbf{a}^T$ is the transpose of $\mathbf{a}$, $\mathbf{C}$ is the systematic parameter covariance matrix of dimension $N_s \times N_s$, n_i^{obs} is the observed number of events in the ith bin, and $n_i^{exp} = n_i^{exp}(\beta;\mathbf{a})$ is the corresponding expected number of events.

The advantage of the likelihood ratio method over the extended maximum-likelihood method is that, in the large-sample limit, the quantity $-2\ln\lambda(\beta;\mathbf{a})$ in Eq.3 has a χ^2 distribution and it can therefore be used as a goodness-of-fit test.

Values of systematic parameters that give a negative predicted number of events in any reconstructed energy bin in any interaction mode are not allowed. If a systematic throw gives a negative number of events, that toy experiment is rejected.

4.2. Marginalization

In the previous analyses a fully frequentist approach, where all the nuisance parameters are profiled, was used. Now a hybrid bayesian-frequentist method is used. The nuisance parameters are marginalized while the parameter of interest is minimized. The marginalization of the nuisances gives a so-called marginal likelihood that depends only on β:

$$\lambda_{marg}(\beta) = \int_A \lambda(\beta;\mathbf{a})\pi(\mathbf{a})d\mathbf{a} = \sum_{i=0}^{n}\lambda(\beta;\mathbf{a}_i)\pi(\mathbf{a}_i) \tag{4}$$

where n is the number of toys, $\mathbf{a}$ is a single extracted set of nuisance parameter, $\pi(\mathbf{a})$ is the prior of the nuisance parameters and $\pi(\mathbf{a}_i)$ is the weight of a particular set of nuisance parameters (e.g. $w = 1/n$ if flat prior). Many sets of $\mathbf{a}$ ($n = 10k$), should be extracted in order to sample the space of the nuisance parameters. The number of extractions depends on the number of nuisance parameters as well as the precision that is required by the measurement.

We derive the marginal log-likelihood, which we will denoted as χ^2 from now on:

$$\chi^2(\beta) = -2\ln\lambda_{marg}(\beta) = -2\ln\left(\sum_{i=0}^{n}\lambda(\beta;\mathbf{a}_i)\pi(\mathbf{a_i})\right) \tag{5}$$

It is important to note that this numerical integration is a weighted average of the likelihood evaluated at the n throws, not a weighted average of the log-likelihood. When marginalization is used the penalty term $(\mathbf{a}-\mathbf{a_0})^T\cdot\mathbf{C}^{-1}\cdot(\mathbf{a}-\mathbf{a_0})$ becomes constant, independent of the parameter of interest. For this reason a modified version of eq. 3, which doesn't include the penalty term, is used.

The toys used for the marginalization of the nuisance parameters are performed exactly in the same way as described in sec. 3.1 , taking into account the correlations, but without statistical fluctuations. First n reconstructed energy distributions (E_{reco}^{marg}) are produced using the randomized systematic parameters and normalized to 1×10^{20} POT, in order to be used for fits at any different number of POT. Finally are they stored in output files which can be used for whatever needed fit. A fit on a single data set is performed as following:

- all the n toy reconstructed energy distributions (E_{reco}^{marg}) are read from the files,
- each E_{reco}^{toy} distribution is normalized to the number of POT of the data set,
- for each E_{reco}^{toy} eq. 3 is calculated and $\lambda(\beta, \mathbf{a}) = e^{-(-2 \ln \lambda(\beta,\mathbf{a}))/2}$ is obtained,
- the χ^2 value is finally obtained, using eq. 5, for a given value of β.

4.3. P-value

There are different test statistics which can be used to quote a p-value, as described in [50] . As decided by the OA group, for this $\bar{\nu}_e$ appearance search two different p-values are quoted. In order to quote the significance given null $\bar{\nu}_e$ appearance, the parameter of interest is set to $\beta = 0$ and $n = 10k$ toys are performed. The systematic parameters are randomized taking into account the correlations as described in sec. 3.1. Then the reconstructed energy distribution is fluctuated with Poisson. Then the following p-values are calculated:

- p-value with rate-only analysis: the number of events is used as test statistics, without using the shape information of the reconstructed energy distribution. The distribution $f(N_{SK})$ of the total number of superK events in a toy (N_{SK}) is obtained from the performed toys. Then the number of events is measured from the data set (N_{SK}^{obs}) and the p-value is computed in the following way:

$$\text{p-value} = \int_{N_{SK}^{obs}}^{+\infty} f(N_{SK}) dN_{SK} = \frac{\sum_{itoy=1}^{n} w(N_{SK} \geq N_{SK}^{obs})^{itoy}}{n} \tag{6}$$

  where $w(N_{SK} \geq N_{SK}^{obs})$ is 1 if $N_{SK} \geq N_{SK}^{obs}$ otherwise 0. The distribution $f(N_{tot}^{toy})$ is expected to be discrete since it follows the Poisson distribution.

- p-value with rate+shape analysis: the shape information of the reconstructed energy distribution is used. In this case the test statistic is used to compare the case of $\beta = 0$ with $\beta = 1$:

$$\Delta\chi^2 = \chi^2(\beta = 0) - \chi^2(\beta = 1) \tag{7}$$

  where $\chi^2(\beta = 0)$ and $\chi^2(\beta = 1)$ are computed using eq. 5. It is worth to note that the parameter β is not fitted but fixed either to 0 or 1. The p-value is calculated as:

$$\text{p-value} = \int_{\Delta\chi^2_{obs}}^{+\infty} f(\Delta\chi^2) d\Delta\chi^2 = \frac{\sum_{itoy=1}^{n} w(\Delta\chi^2 \geq \Delta\chi^2_{obs})^{itoy}}{n} \tag{8}$$

  where $w(\Delta\chi^2 \geq \Delta\chi^2_{obs})$ is 1 if $\Delta\chi^2 \geq \Delta\chi^2_{obs}$ otherwise 0.

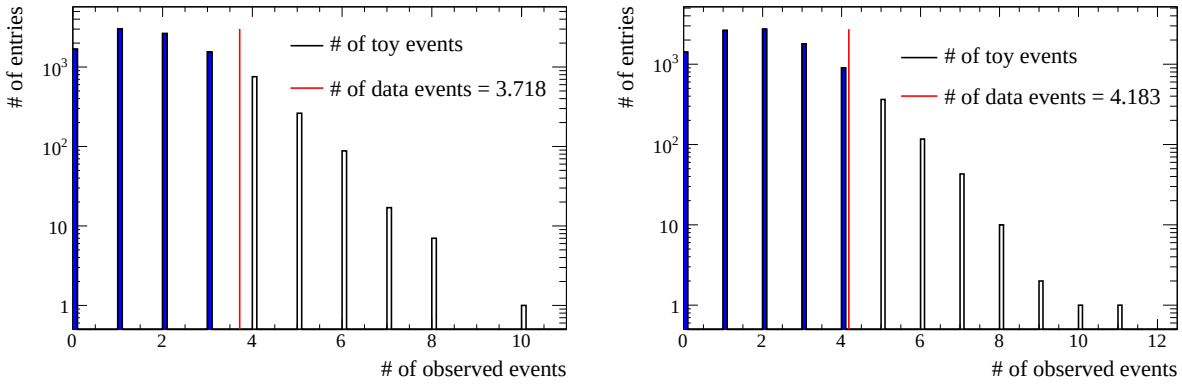


Figure 25: The distribution of the total number of events from each of the 10k toy experiments. The red line corresponds to the total number of events given by the Asimov dataset. The study is done with 4.0×10^{20}POT (left) and 4.5×10^{20}(right).

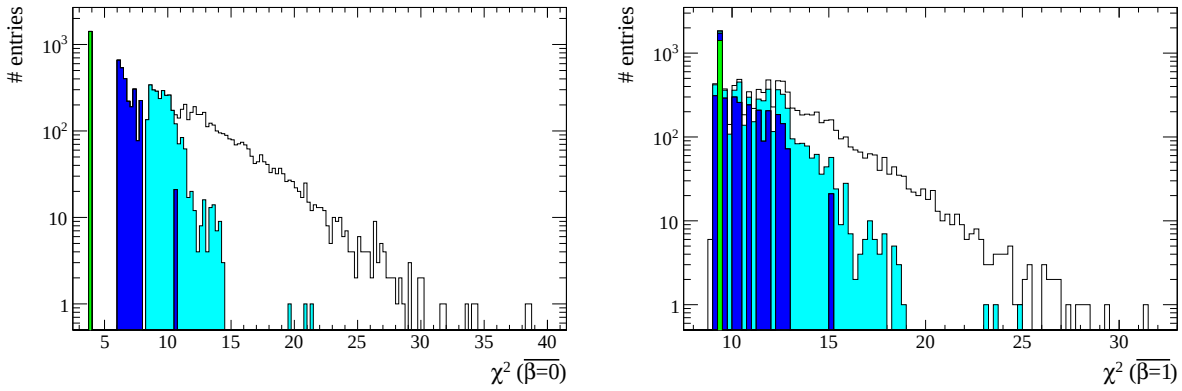


Figure 26: Distribution of the χ^2 for $\beta = 0$ (left) and $\beta = 1$ (right). They correspond to 4.5×10^{20}POT. The colored green entries correspond to the toys with no observed events while the blu and the cyan entries correspond respectively to toys with 1 and 2 events.

5. Sensitivity to $\bar{\nu}_e$ appearance

As discussed within the OA group, since there are not fitted parameters in the analysis, the validation can be done by comparing the rate-only with rate+shape p-values and checking if the corresponding distributions are reasonable. This is done by performing a sensitivity study to $\bar{\nu}_e$ appearance both with the Asimov dataset, which consists of the most probable dataset that is the nominal expected reconstructed energy distribution [42], as well as for toy distributions, give a value of β.

The significance to $\bar{\nu}_e$ appearance is quoted for both $4.0\times10^{20}$and 4.5×10^{20}POT using the procedure described in sec. 4.3.

5.1. Median rate-only p-value

The expected median rate-only p-value is given by the Asimov dataset for $\beta = 1$. The corresponding oscillation parameters are shown in table 6, The distribution of the number of superK events used as test statistic is shown in figure 25. The number of events is 3.718 for 4.0×10^{20}POT and 4.183 for 4.5×10^{20}POT. The quoted p-value is and 0.113 for 4.0×10^{20}POT. and 0.0538 for 4.5×10^{20}POT.

5.2. Median rate+shape p-value

The χ^2 distributions for both $\beta = 0$ and $\beta = 1$ are shown in figure 27. They don't follow a real χ^2 distribution because the number of events is low, as explained in sec. 4.1. In particular it is visible that the region where spikes are present is dominated by a low number of events, between 0 and 2. Since the

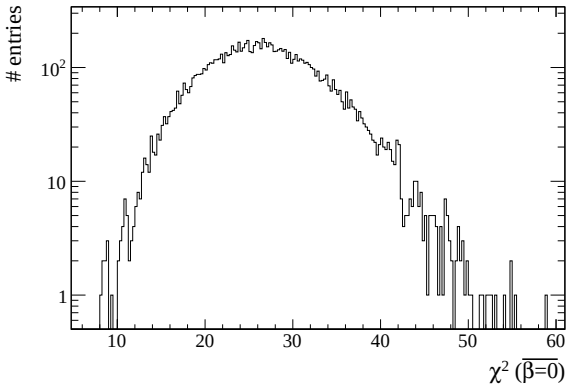


Figure 27: Distribution of the χ^2 for $\beta = 0$. They correspond to 7.8×10^{21}POT.

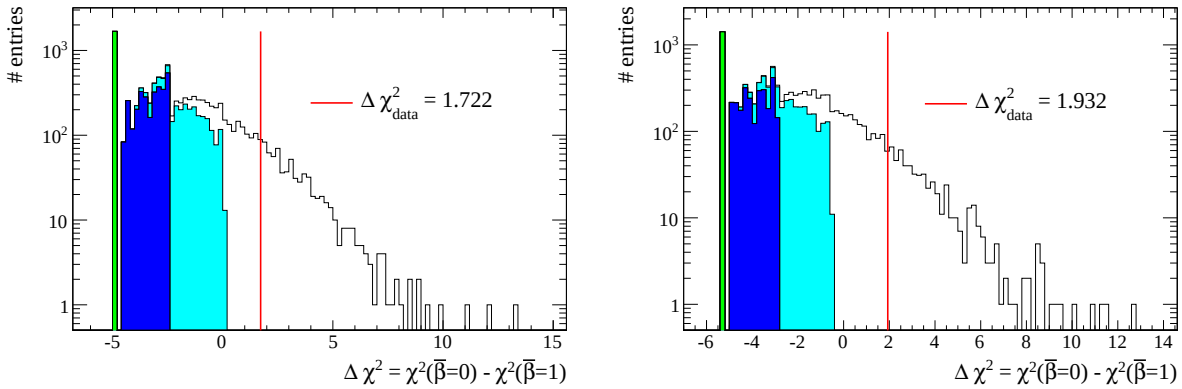


Figure 28: Distribution of the $\Delta\chi^2$ for each of the 10k toy experiment. The red line corresponds to $\Delta\chi^2$ obtained from the Asimov dataset. The study is done with 4.0×10^{20}POT (left) and 4.5×10^{20}(right). The colored green entries correspond to the toys with no observed events while the blu and the cyan entries correspond respectively to toys with 1 and 2 events.

statistical uncertainty is very large and dominant, the shape information cannot be fully exploited and the distributions partially show a discrete behavior. Indeed if the statistics is increased to the expected full T2K number of POT, 7.8×10^{21}, the distributions behave as a χ^2 function with 25 degrees of freedom (the number of reconstructed energy bins), as expected.

The expected distribution of $\Delta\chi^2$ for all the 10k toy experiments is shown in Fig. 28 for both $4.0\times10^{20}$and 4.5×10^{20}POT. The $\Delta\chi^2$ from the Asimov dataset is 1.932 with 4.5×10^{20}POT, while for 4.0×10^{20}it is 1.72185. The median expected p-values are 0.0723 for 4.0×10^{20}POT and 0.0572 for 4.5×10^{20}POT. The rate-only p-value is smaller than the rate+shape p-value because of the discrete behaviour of the test statistic in the rate-only analysis. This feature is discussed more in detail in Sec. 5.3.

The p-value for the Asimov dataset as a function of β or the total number of events is shown for 4.5×10^{20}respectively in fig. 31 and fig. 32.

5.3. Mean p-value

The mean p-value is computed without using the Asimov dataset, but with toy experiments:

- the distributions shown in Fig. 25 and Fig. 28 are used respectively for the rate-only and rate+shape p-value computations,
- 10k toys are produced with $\beta = 1$,
- for each toy with $\beta = 1$ the test statistics (i.e. the number of events and $\Delta\chi^2$ given in Eq. 7) are computed and a p-value is quoted using the distributions Fig. 25 and Fig. 28,

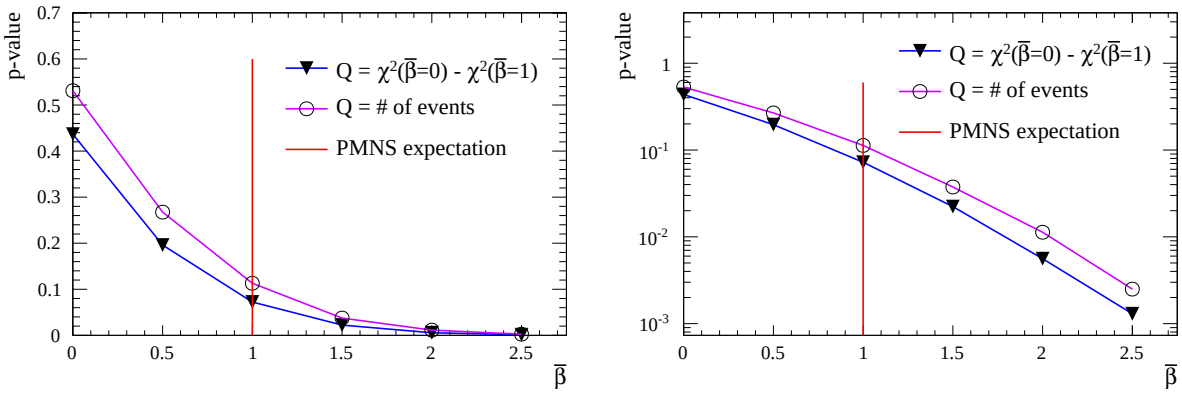


Figure 29: Rate-only (violet) and rate+shape (blue) p-value given by Asimov datasets for several values of β in normal (up) and logarithmic (bottom) scale. The red line corresponds to the PMNS expectation ($\beta = 1$). The study is done with 4.0×10^{20}POT.

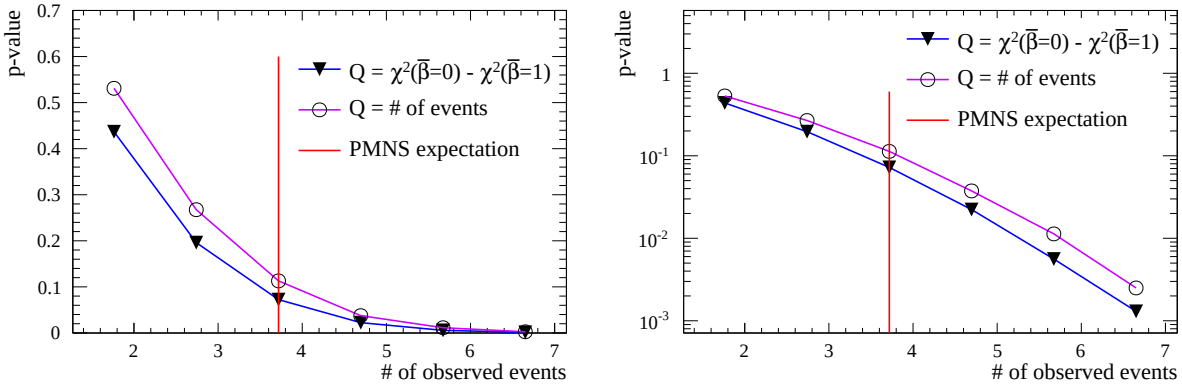


Figure 30: Rate-only (violet) and rate+shape (blue) p-value for different number of observed events given by Asimov datasets corresponding to the several values of β shown in figure 29. Both normal (up) and logarithmic scales (bottom) are shown. The red line corresponds to the PMNS expectation ($\beta = 1$). The study is done with 4.0×10^{20}POT.

- a distribution of rate-only and rate+shape p-values is obtained and the mean p-value is calculated.

The mean rate-only p-value, given by the mean of the p-value distribution shown in Fig. 33, is 0.209 while the mean rate+shape p-value is 0.143 for 4.0×10^{20}POT. For 4.5×10^{20}POT the rate-only p-value is 0.188 while the rate+shape is 0.127. For both cases the mean p-value is larger than the median one, which corresponds to the most probable p-value. It's worth to point out that unlike for the mean p-value obtained with the Asimov dataset, the mean rate-only p-value is smaller than the mean rate+shape p-value. It means that, as expected, on average the rate+shape p-value is smaller than the rate-only one, since it exploit also the shape information. However if a single dataset is used, like the Asimov one, it can happen that, due to the discreteness of test statistic (number of events), the rate-only p-value is smaller than the rate+shape p-value. Indeed a change of the number of events produce a not small jump in the p-value, as shown in the left plot of Fig. 33, while the distribution of the rate+shape p-value is smooth and continuos. For this reason it is important to quote both the p-values.

In Fig. 34 the rate+shape p-value as a function of the number of events for single toy produced with $\beta = 1$ is shown for 4.5×10^{20}POT.

5.4. Summary of expected significance to $\bar{\nu}_e$ appearance

In table 11 and table 12 the expected rate-only and rate+shape p-values corresponding respectively to 4.0×10^{20}POT and 4.5×10^{20}POT are shown.

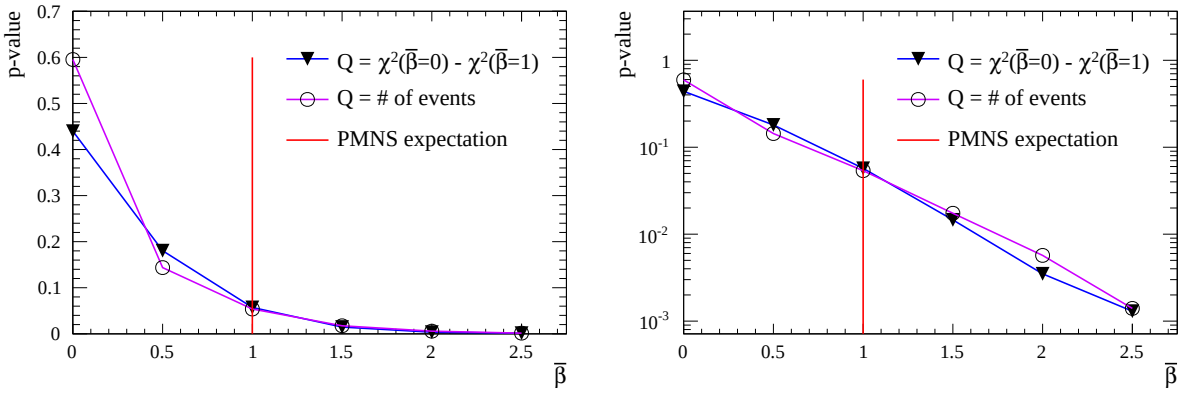


Figure 31: Rate-only (violet) and rate+shape (blue) p-value given by Asimov datasets for several values of β in normal (up) and logarithmic (bottom) scale. The red line corresponds to the PMNS expectation ($\beta = 1$). The study is done with 4.5×10^{20}POT.

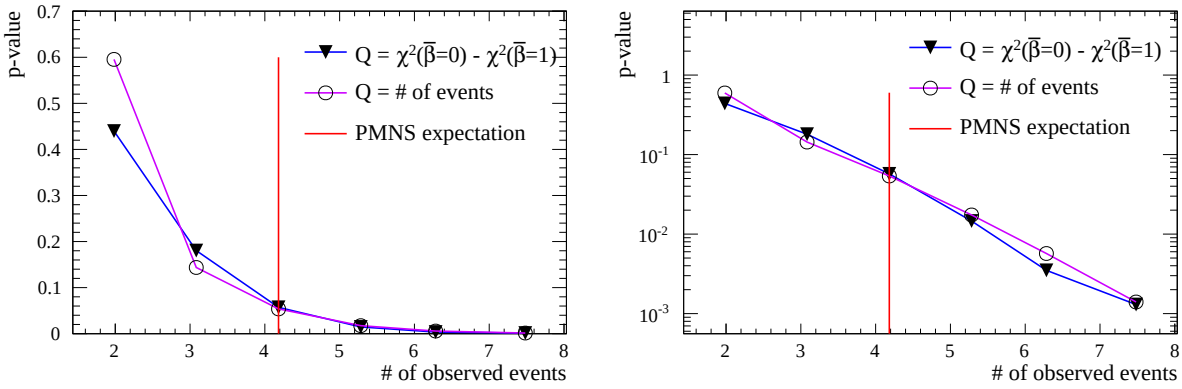


Figure 32: Rate-only (violet) and rate+shape (blue) p-value for different number of observed events given by Asimov datasets corresponding to the several values of β shown in 31. Both normal (up) and logarithmic scales (bottom) are shown. The red line corresponds to the PMNS expectation ($\beta = 1$). The study is done with 4.5×10^{20}POT.

P-value	Median	Mean
Rate-only	0.113	0.209
Rate+shape	0.0723	0.143

Table 11: Summary of the expected rate-only and rate+shape p-values for 4.0×10^{20}POT. The median p-value is calculated using the Asimov dataset while the mean p-value is calculated making 10k toys and taking the mean of the p-value distribution.

P-value	Median	Mean
Rate-only	0.0538	0.188
Rate+shape	0.0572	0.127

Table 12: Summary of the expected rate-only and rate+shape p-values for 4.5×10^{20}POT. The median p-value is calculated using the Asimov dataset while the mean p-value is calculated making 10k toys and taking the mean of the p-value distribution.

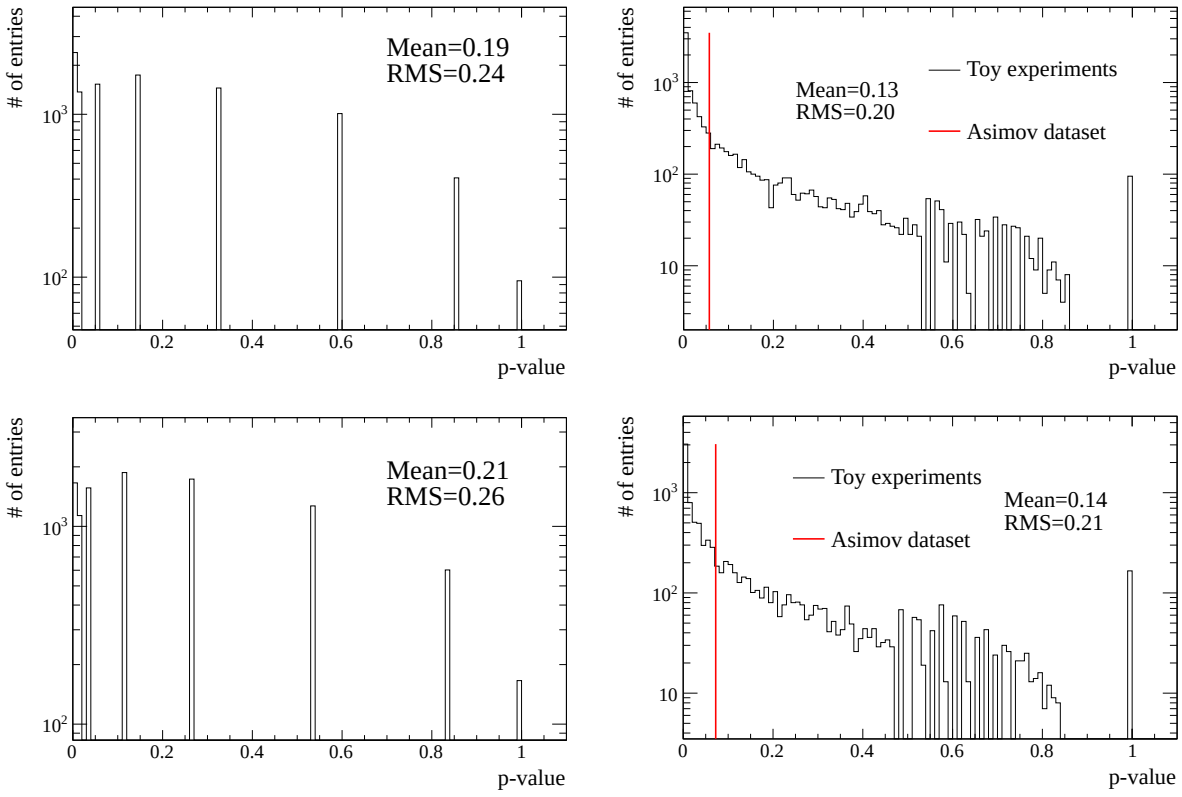


Figure 33: Distribution of rate-only (left) rate+shape (right) p-values obtained by 10k toys with $\beta = 1$. The rate-only p-value is discrete and each filled bin correspond to toys with different number of events. In both distributions the spike at p-value=1 corresponds to toys with no events. The study is performed with 4.0×10^{20}(up) 4.5×10^{20}(down) POT.

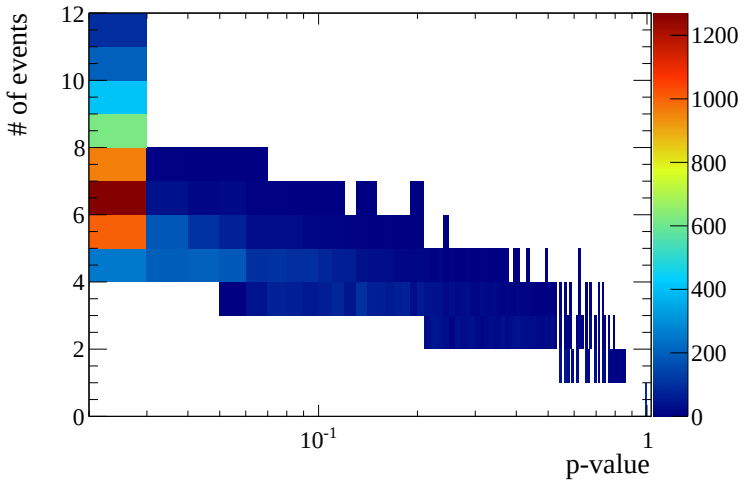


Figure 34: The rate+shape p-value as a function of the number of events for a single toy produced with $\beta = 1$ is shown for 4.5×10^{20}POT. 10k toys are produced as described in Sec. 5.3. Each entry corresponds to a different toy.

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Figure 35: Distribution of the $\Delta\chi^2$ for each of the 10k toy experiment. The red line corresponds to $\Delta\chi^2$ obtained from the data.

6. Results of the $\bar{\nu}_\mu$-disappearance analysis with the Run 5+6 dataset

6.1. Significance to $\bar{\nu}_e$ appearance

The rate-only and rate+shape p-values for $\bar{\nu}_e$-appearance are performed on the combined 4.5×10^{20} POT Run 5+6 dataset following the same procedure as in section 5. All 56 systematic parameters are marginalized. All the oscillation parameters except $\sin^2\theta_{12}$ and Δm^2_{21}, as well as the mass hierarchy, are marginalized.

A summary of the observed number of events in SuperK, $\Delta\chi^2$ and p-values is given in table 28.

Observed num. of events	p-value (rate-only)	$\chi^2(\beta=0)$	$\chi^2(\beta=1)$	$\Delta\chi^2$	p-value (rate+shape)
XXX	XXX	XXX	XXX	XXX	XXX

Table 13: Summary of observed number of events, χ^2 for $\beta=0$ and $\beta=1$, $\Delta\chi^2$, rate-only and rate+shape p-values from the combined Run 5+6 dataset. The quoted χ^2 values were computed from Eq. 5, while $\Delta\chi^2$ from eq. 7 in Sec. 4.1. using the reconstructed energy binning scheme given in Sec. 2.2.5.

7. Summary

In this technical note, we have presented the results of a 3-flavour $\bar{\nu}_e$-appearance analysis, performed by the *VALOR* group, on the combined Run 5+6 4.5×10^{20} dataset, using only data taken in antineutrino beam mode. The goal of this analysis is to search for $\bar{\nu}_e$ appearance in a $\bar{\nu}_\mu$ beam. This analysis has adapted the work presented in Refs. [1], [2], [3], [4], [5] and [6] to this dataset. The Run 5+6 dataset corresponds to an integrated J-PARC neutrino beam exposure of 4.5×10^{20} POT. This analysis predicts 4.183 $\pm$ 0.734 (syst) single e-like ring events in Super-K, if $\bar{\nu}_e$ appearance is considered, while XXX were observed. A $\bar{\nu}_e$-appearance analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. The observed reconstructed energy spectrum of single e-like ring events was used to compute the p-values. A rate-only p-value is performed using only the total number of observed events, while a binned log-likelihood ratio was used to compute the rate+shape p-value, with a test statistic, $\Delta\chi^2$, which compares the case for $\beta = 0$ and $\beta = 1$. Also all 56 systematic parameters considered in this analysis were marginalized. Among them all the oscillation parameters, except $sin^2\theta_{12}$ and Δm^{21}, as well as the mass hierarchy were marginalized. The quote rate-only p-value is XXX, while the rate+shape p-value is XXX.

References

[1] C.Andreopoulos, T.Dealtry, J.Dobson, N.Grant, M.Haigh, J.Ilic, G.Pearce and A.Weber, T2K 3.23×10^{19}-POT Muon-Neutrino Disappearance Analysis, T2K-TN-036(v4), http://www.t2k.org/docs/technotes/036

[2] C.Andreopoulos, T.Dealtry, J.Dobson, N.Grant, M.Haigh, J.Ilic, G.Pearce and A.Weber, T2K 1.431×10^{20}-POT Muon-Neutrino Disappearance Analysis T2K-TN-064(v3), http://www.t2k.org/docs/technotes/064

[3] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K 1.431×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-087(v1), http://www.t2k.org/docs/technotes/087

[4] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K 3.010×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-141(v10), http://www.t2k.org/docs/technotes/141

[5] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K 6.3933×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-155(v2), http://www.t2k.org/docs/technotes/155

[6] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K 6.57×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-183(v3), http://www.t2k.org/docs/technotes/183

[7] C.Andreopoulos et al., T2K 3.010×10^{20}-POT Joint 3-Flavour Oscillation Analysis T2K-TN-154(v3), http://www.t2k.org/docs/technotes/154

[8] C.Andreopoulos et al., T2K 6.57×10^{20}-POT Joint 3-Flavour Oscillation Analysis T2K-TN-175, http://www.t2k.org/docs/technotes/175

[9] C.Andreopoulos et al., T2K 2.315×10^{20}-POT 3-Flavour Muon-Antineutrino Disappearance Analysis T2K-TN-243, http://www.t2k.org/docs/technotes/243

[10] C.Andreopoulos, T.Dealtry, L.Escudero and N.Grant, T2K 6.5700×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis: Supplementary material on the effect of systematics, T2K-TN-183 supplement, http://www.t2k.org/docs/technotes/183

[11] T2K Collaboration (K.Abe et al.), First Muon-Neutrino Disappearance Study with an Off-Axis Beam, Phys.Rev.D85:031103, 2012; arXiv:1201.1386

[12] T2K Collaboration (K.Abe et al.), Measurement of Neutrino Oscillation Parameters from Muon Neutrino Disappearance with an Off-axis Beam, arXiv:1308.0465 [hep-ex], 2013.

[13] T2K Collaboration (K.Abe et al.), Measurement of neutrino oscillation in appearance and disappearance channels by the T2K experiment with 6.6×10^{20} protons on target http://www.t2k.org/comm/pubboard/JournalPapers/oa2014prd/draft3

[14] Y.Hayato, A neutrino interaction simulation program library NEUT, Acta Phys.Polon.B40:2477-2489,2009.

[15] Beam WG, Flux tuning (11b v3.2), http://www.t2k.org/beam/NuFlux/FluxRelease/11brelease/11btuned-v3.2

[16] M.Otani and J.Kameda, Super-K events and Updated systematic errors for ν_μ disappearance analysis with T2K 1.431×10^{20} POT T2K-TN-065(v6), http://www.t2k.org/docs/technotes/065

[17] P. de Perio et al., NEUT nuclear effects (FSI) T2K-TN-033(v2.0), http://www.t2k.org/docs/technotes/033

[18] P. de Perio et al., Cross section parameters for the 2012a oscillation analysis T2K-TN-108(v1.3), http://www.t2k.org/docs/technotes/108

[19] P. de Perio et al., Implementation of the NIWG cross-section parametrization, T2K-TN-113(v1.2)

[20] Y.Hayato, K.McFarland, and M.Wascko (Private communication)

[21] P. de Perio et al., Pion Hadronic Secondary Interactions in Super-Kamiokande, T2K-TN-105(v1.1) http://www.t2k.org/docs/technotes/105

[22] Y. Ashie et al., Phys. Rev. Lett. 93, 101801 (2004); Y. Ashie et al., Phys. Rev. D71, 112005 (2005); J. Hosaka et al., Phys. Rev. D 74, 032002 (2006).

[23] Beam group, http://www.t2k.org/beam/NuFlux/FluxRelease/11arelease/11atuned-v2.3.

[24] M. Wilking, Status of the 2013 ND280 ν_μ analysis, http://www.t2k.org/asg/meeting/20130411/nd280numustatus/view.

[25] M. Hartz, A. Kaboth, and K. Mahn, Constraining the Flux and Cross Section Models with Data from the ND280 Detector for the 2013 Oscillation Analysis, T2K-TN-166(v3.0) http://www.t2k.org/docs/technotes/166.

[26] M. Hartz, A. Kaboth, K. Mahn, Jordan Myslik, Christine Nielsen and Mark Scott, Constraining the Flux and Cross Section Models with Data from the ND280 Detector for the 2014/2015 Oscillation Analysis, T2K-TN-220(v1.0) http://www.t2k.org/docs/technotes/220.

[27] J. Kameda, Updated study of the systematic error in ν_μ disappearance analysis from Super-Kamiokande, T2K-TN-159(v2), http://www.t2k.org/docs/technotes/159.

[28] J. Hignight et al., Super-Kamiokande events and data quality studies for T2K Run4 T2K-TN-148(v2r3), http://www.t2k.org/docs/technotes/148.

[29] 14a data and MC info http://www.t2k.org/t2ksk/documentation-page/MCWeighting/MCReweighting14a

[30] 2015 SK splines http://www.t2k.org/asg/oagroup/inputs/201415/2015SKsplines

[31] SuperK 13a flux release http://www.t2k.org/beam/NuFlux/FluxRelease/13arelease

[32] A. Bercellie et al. Cross section parameters for 2014 oscillation analysis. T2K-TN-192 (v2.1) http://www.t2k.org/docs/technotes/192/ .

[33] T. Wongjirad, A. Minamino and J. Kameda, Muon neutrino disappearance analysis with T2K 6.57 $\times$ 10^{20} POT data, T2K-TN-164(v4), http://www.t2k.org/docs/technotes/164

[34] C. Bronner et al., Run1-4 p-theta electron neutrino appearance analysis, T2K-TN-161(v2.6), http://www.t2k.org/docs/technotes/161

[35] Y. Nishimura, J. Hignight, and S. Nakayama ν_e appearance analysis of RUN1-4 data with neutrino energy spectrum, T2K-TN-162(v2.2), http://www.t2k.org/docs/technotes/162

[36] A. Kaboth, R. Calland and D. Payne (Mach3) A Joint ND280-SK 1R$_\mu$-SK 1R$_e$ Fit using MCMC T2K-TN-171 http://www.t2k.org/docs/technotes/162

[37] N Abgrall et al. Measurements of production properties of K^0_S mesons and Λ hyperons in proton-carbon interactions at 31 GeV/c Phys. Rev. C 89, 025205

[38] N Abgrall et al. Measurements of Cross Sections and Charged Pion Spectra in Proton-Carbon Interactions at 31 GeV/c. Phys.Rev., C84:034604, 2011; N. Abgrall et al. Measurement of production properties of positively charged kaons in proton-carbon interactions at 31 GeV/c. arXiv:hep-ex/1112.0150, 2011.

[39] J. Beringer et al. (Particle Data Group), Phys. Rev. D86, 010001 (2012)

[40] Lorena Escudero Sanchez for the *VALOR* group, http://www.t2k.org/asg/oagroup/meeting/20121219/VaLOR

[41] Tom Dealtry for the *VALOR* group, http://www.t2k.org/asg/meeting/2012/20121015/valornumudisapp/view

[42] G.Cowan, K.Cranmer, E.Gross, O.Vitells, "Asymptotic formulae for likelihood-based tests of new physics" arXiv:1007.1727v2

[43] G. Cowan (1998) Statistical data analysis, Oxford University Press, ISBN 0-19-850155-2.

[44] F.James, M.Roos, Minuit: A System for Function Minimization and Analysis of the Parameter Errors and Correlations, Comput.Phys.Commun. 10:343 (1975)

[45] G.J. Feldman and R.D. Cousins, A Unified approach to the classical statistical analysis of small signals, Phys.Rev.D57:3873-3889,1998; physics/9711021

[46] R.D. Cousins and V.L.Highland, Nucl.Instr. and Meth. A320:331035, 1992.

[47] J. Stuart, A. Ord and S.Arnold, Kendall's Advanced Theory of Statistics, Vol 2A (6th Ed.) (Oxford University Press, New York, 1994).

[48] N.Grant for the *VALOR* group, http://www.t2k.org/asg/meeting/2011/20111124/pvalues/view

[49] D.Karlen, p-values, distributions and low statistics, http://www.t2k.org/asg/oagroup/meeting/20120416/lowstat/view

[50] D.Sgalaberna for the *VALOR* group, http://www.t2k.org/asg/meeting/2015-05-14/view

[51] K. McFarland for the T2K ν_μ-disappearance contour committee (C.Andreopoulos, N.Grant, J.Kameda, D.Karlen, E.Kearns, K.McFarland, M.Otani, M.Zito) Report to ASG conveners meeting, 17 May 2011, http://www.t2k.org/asg/contour/ASG-Conveners-Report/view

[52] R.Nichol, Talk at Neutrino 2012

[53] P. Adamson, et al., Measurement of Neutrino and Antineutrino Oscillations Using Beam and Atmospheric Data in MINOS, arXiv:1304.6335 [hep-ex], 2013

[54] A. Radovic, Talk at DPF 2013

[55] Y. Itow, Recent results in atmospheric neutrino oscillations in the light of large θ_{13}, Nucl. Phys. B (Proc. Suppl.) 235 79 (2013).

[56] G.L.Fogli, E.Lisi et al., Evidence of theta(13)>0 from global neutrino data analysis, Phys.Rev.D84:053007, 2011; arXiv:1106.6028 [hep-ph]

[57] M.C.Gonzalez-Garcia, M.Maltoni, J.Salvado and T.Schwetz, Global fit to three neutrino mixing: critical look at present precision, arXiv:1209.3023 [hep-ph]

[58] Prob3++ 3-flavour oscillation probability software, http://www.phy.duke.edu/ raw22/public/Prob3++/

[59] P. Adamson *et al.* [MINOS Collaboration], Phys. Rev. Lett. **110** (2013) 25, 251801 [arXiv:1304.6335 [hep-ex]].